

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-13 17:41:37

Data & configuration

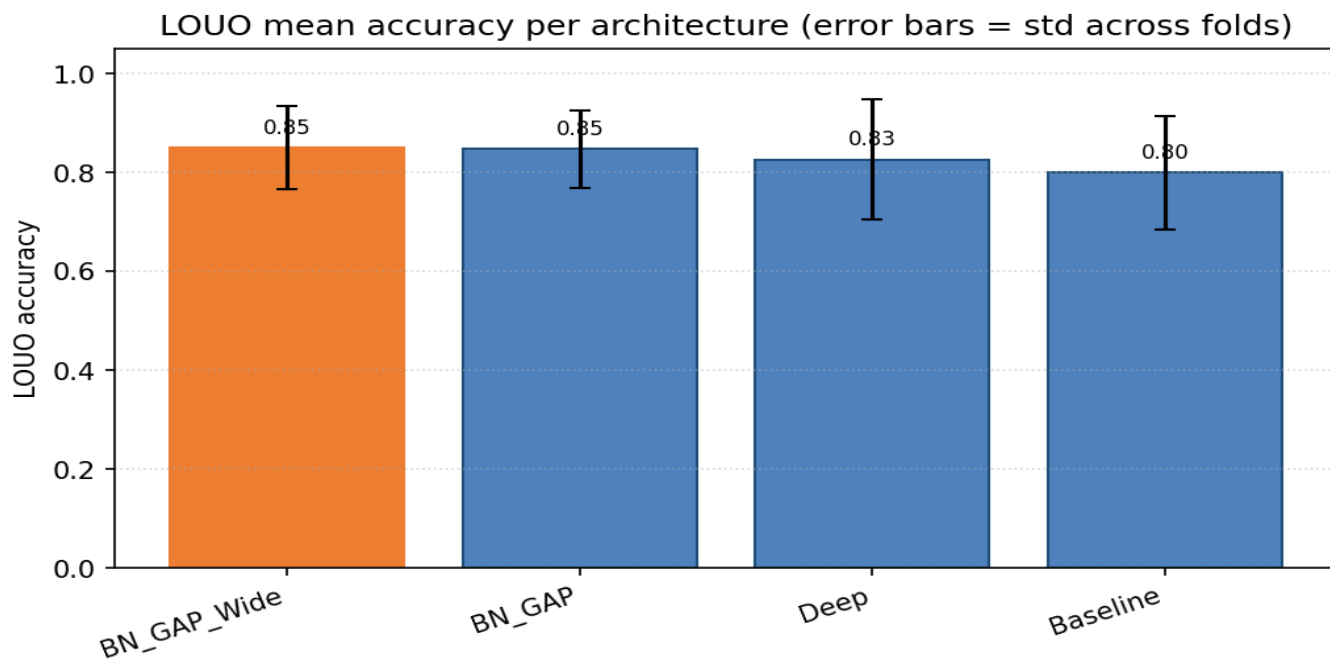
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Static
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Static
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Static
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Static
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/5_Georgia/Static
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Static
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/7_Henry/Static
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/9_Joselyn/Static
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/10_Josh/Static
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Static
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Static
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/13_Nat/Static
LOUO path 14	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Static
LOUO path 15	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Static
LOUO path 16	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Static
LOUO users loaded	16 (10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn)
Excluded classes	Double_Nothing
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	off
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	BN_GAP_Wide, Deep, Baseline, BN_GAP
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

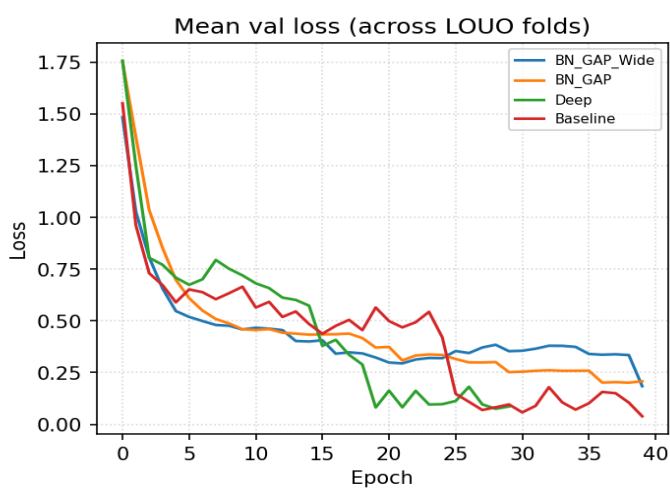
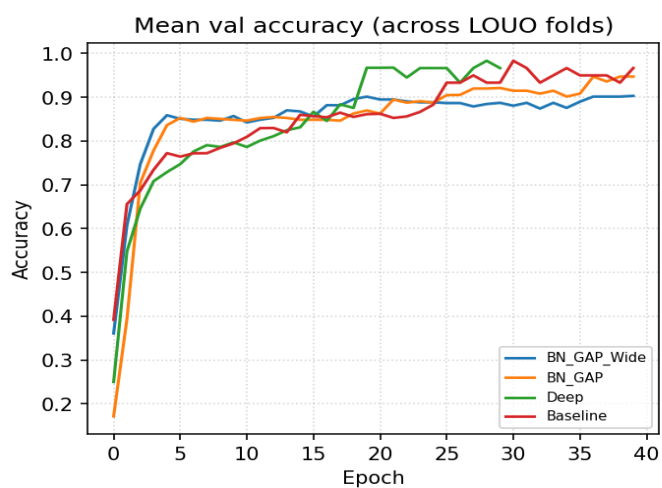
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
BN_GAP_Wide	104,294	0.8493	0.0834	16	—
BN_GAP	57,190	0.8465	0.0778	16	—
Deep	203,878	0.8255	0.1215	16	—
Baseline	113,190	0.7994	0.1147	16	—

Best LOUO mean: BN_GAP_Wide

LOUO accuracy per variant

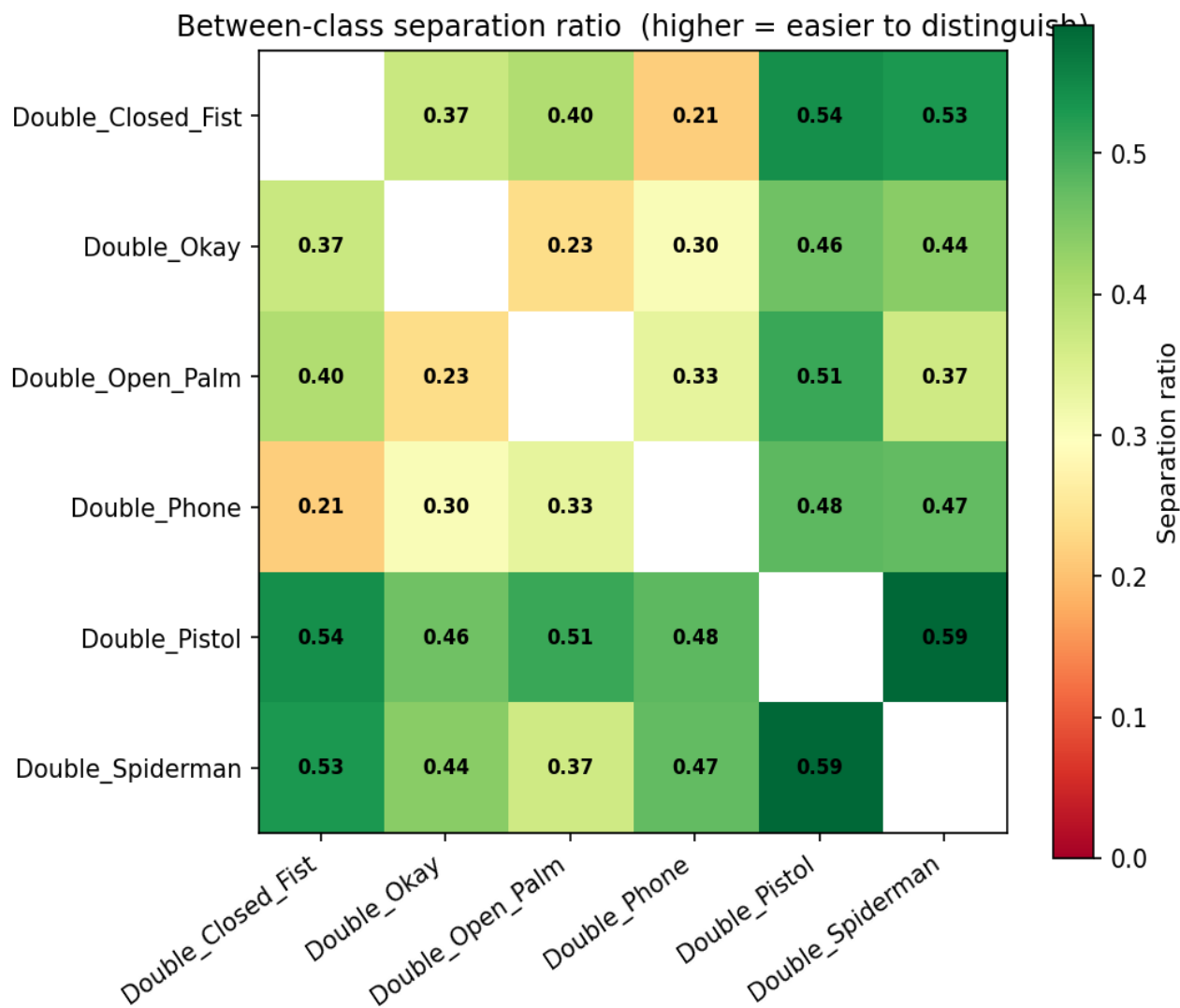
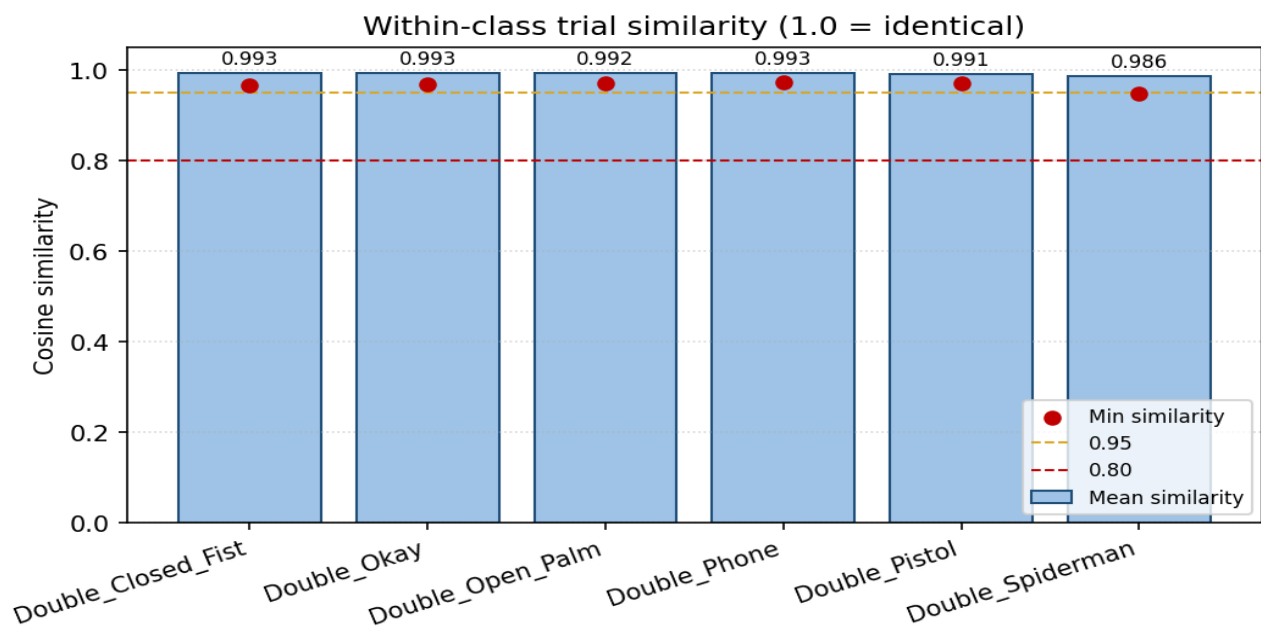


Mean training curves across LOUO folds

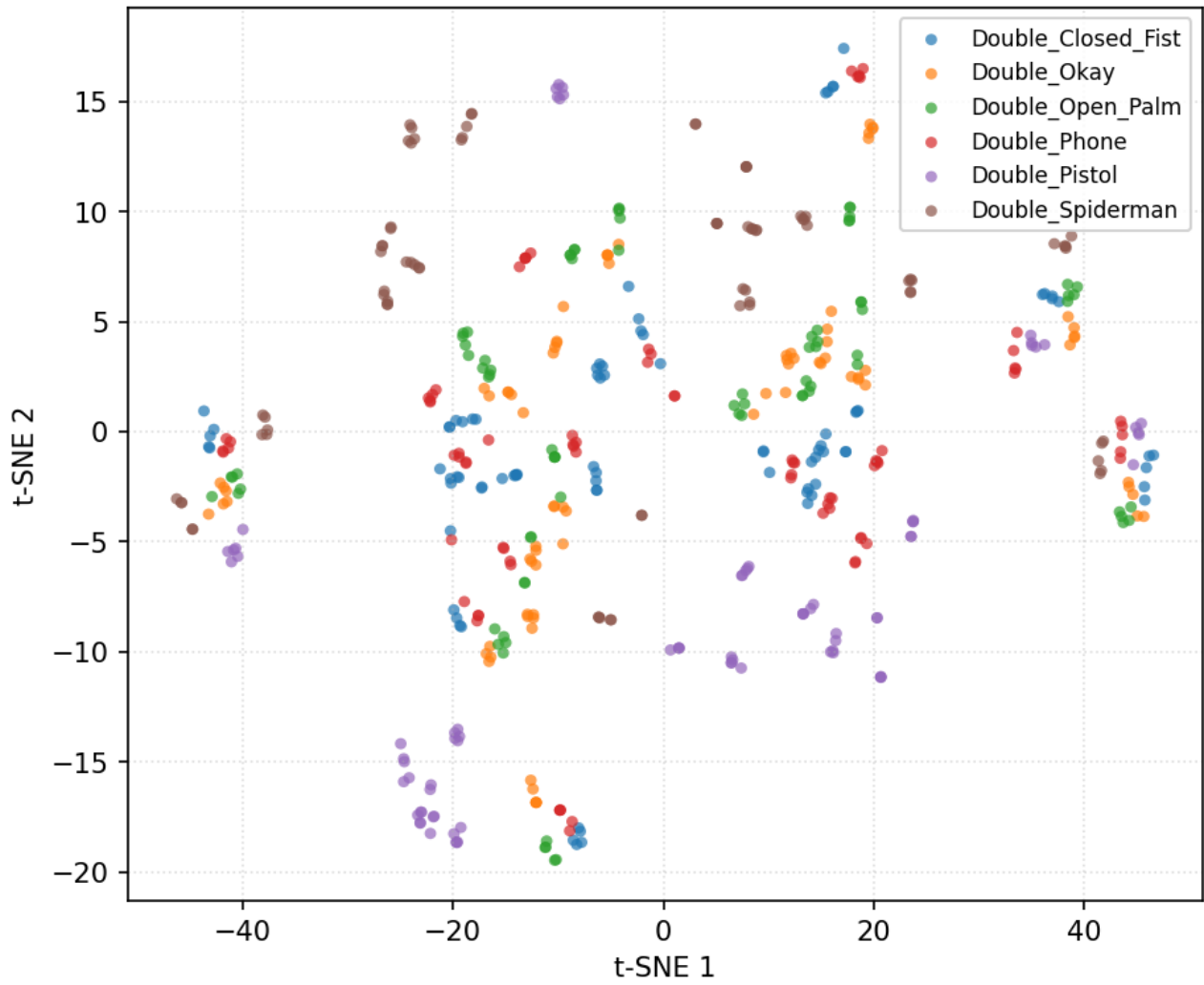


Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).



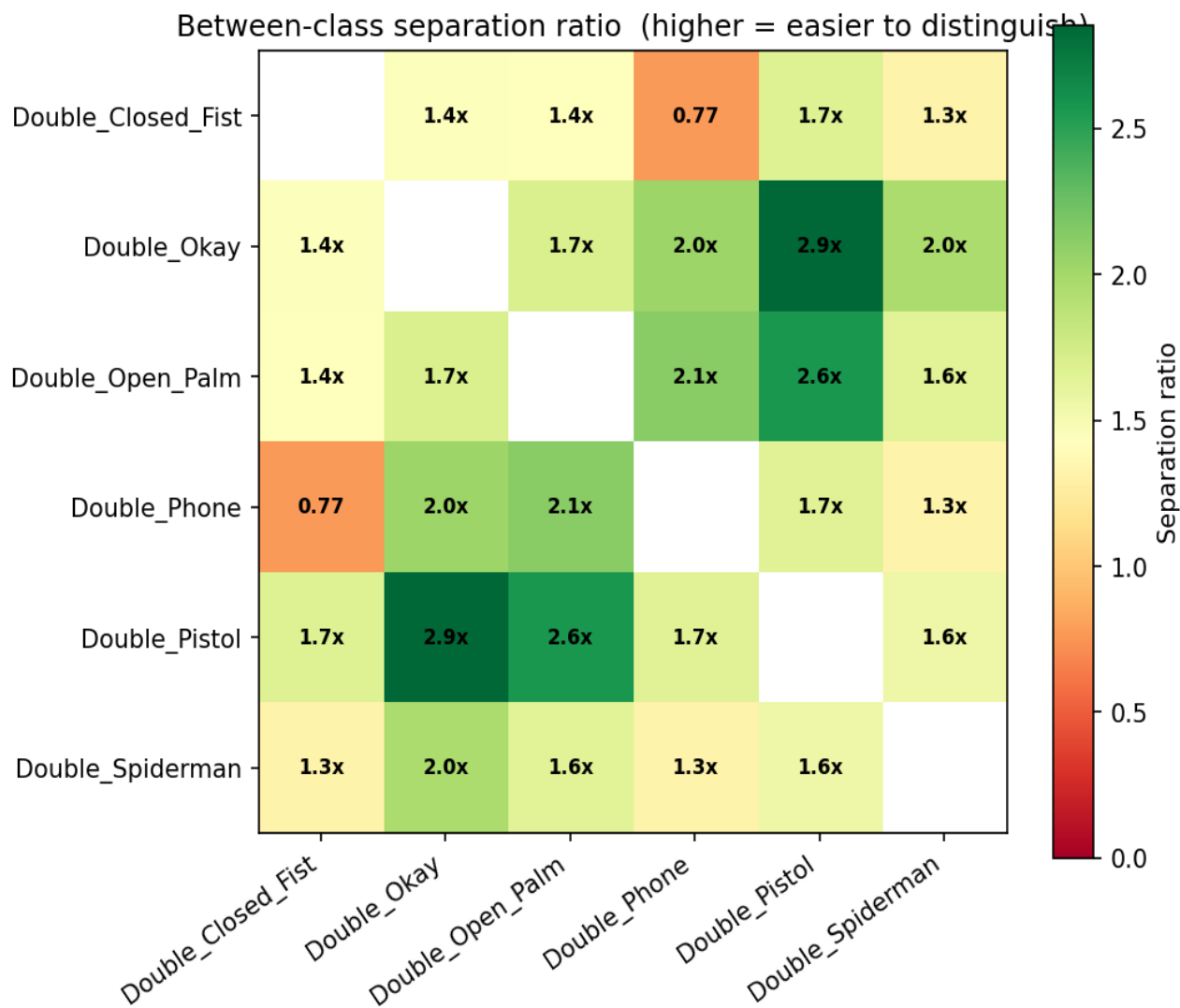
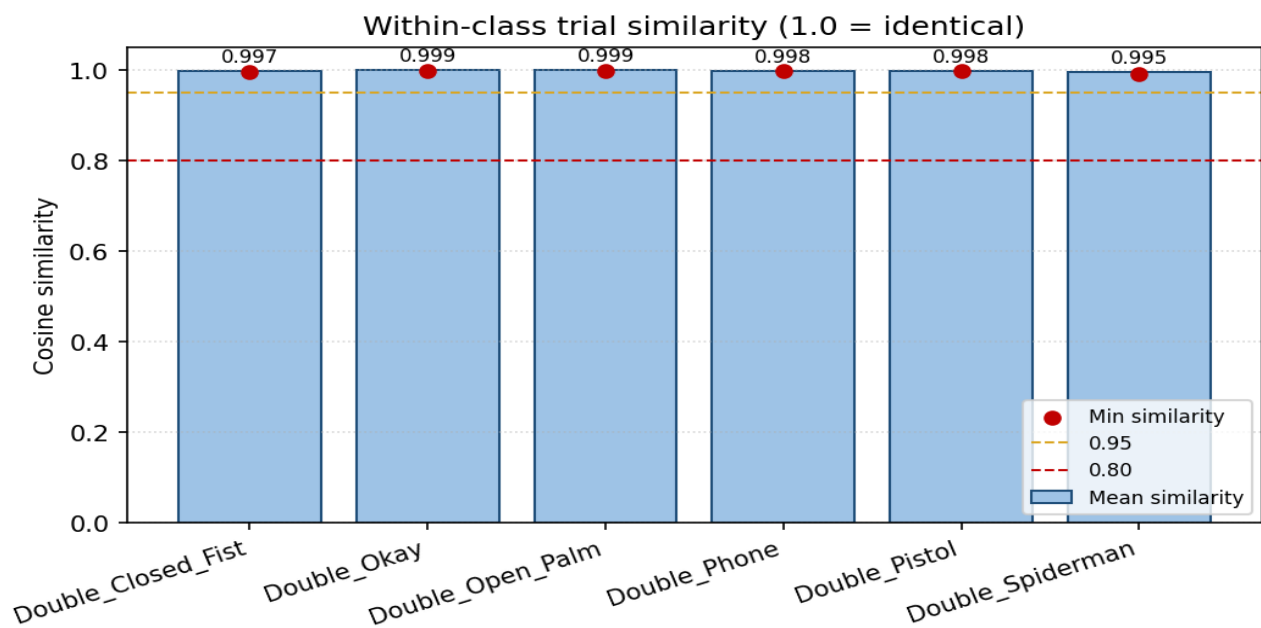
t-SNE feature space projection



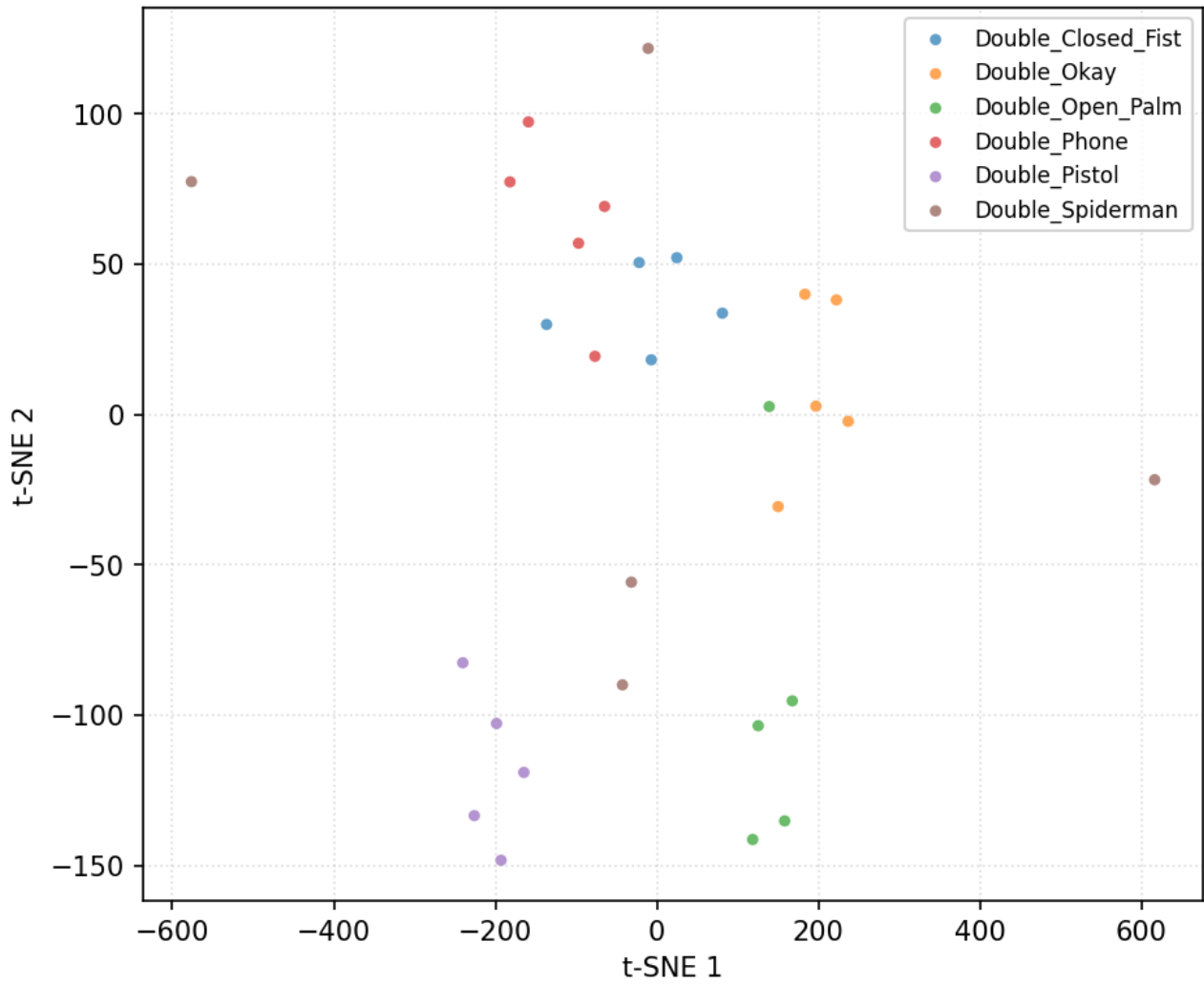
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 10_Josh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

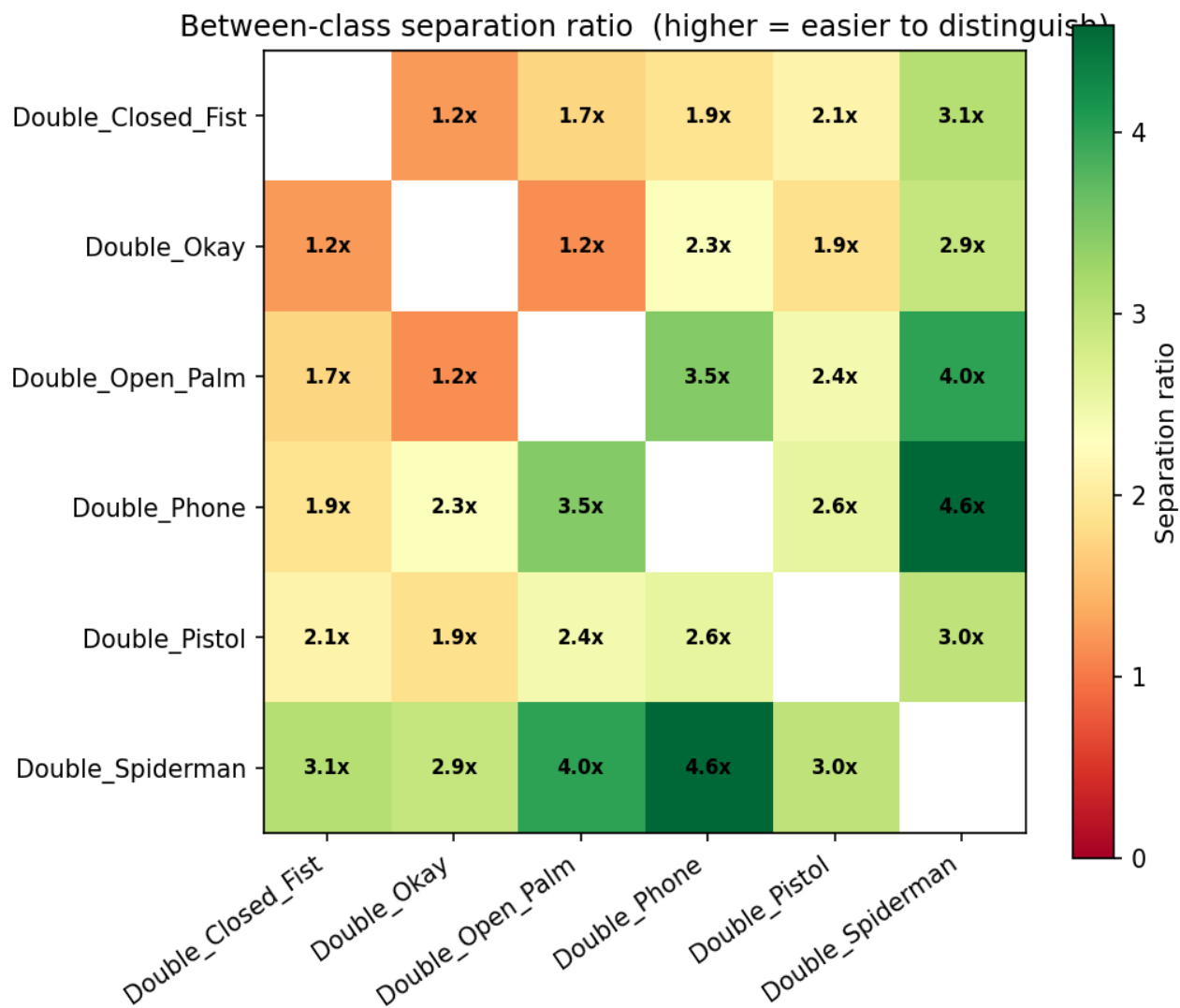
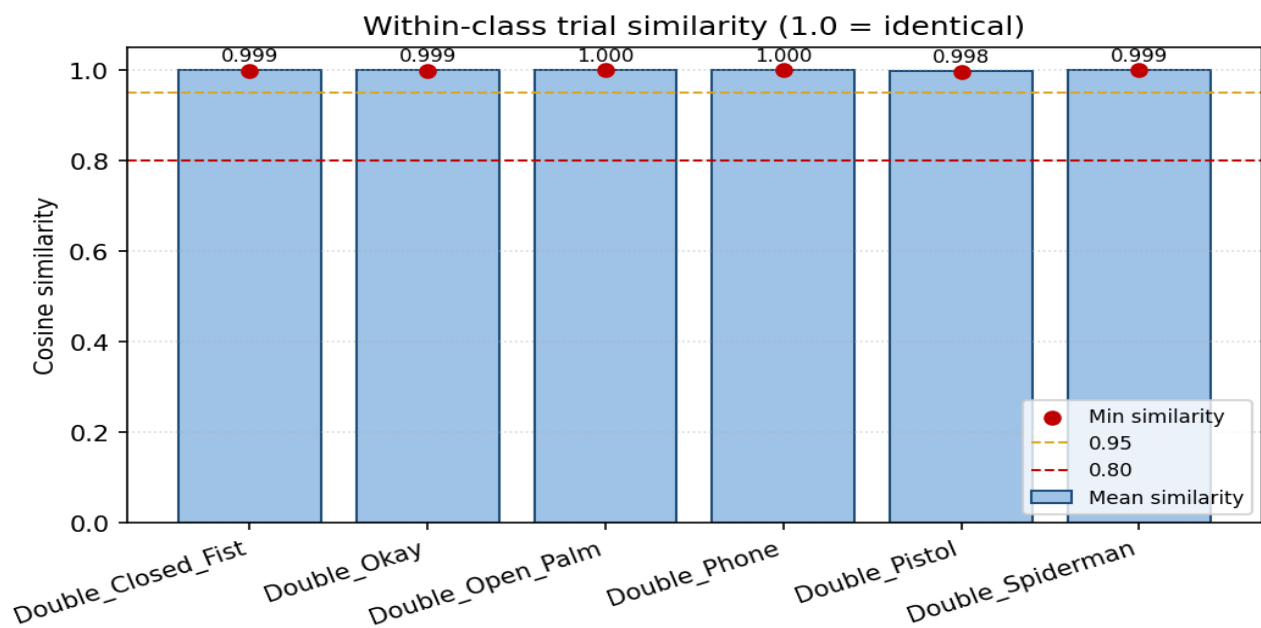


t-SNE feature space projection

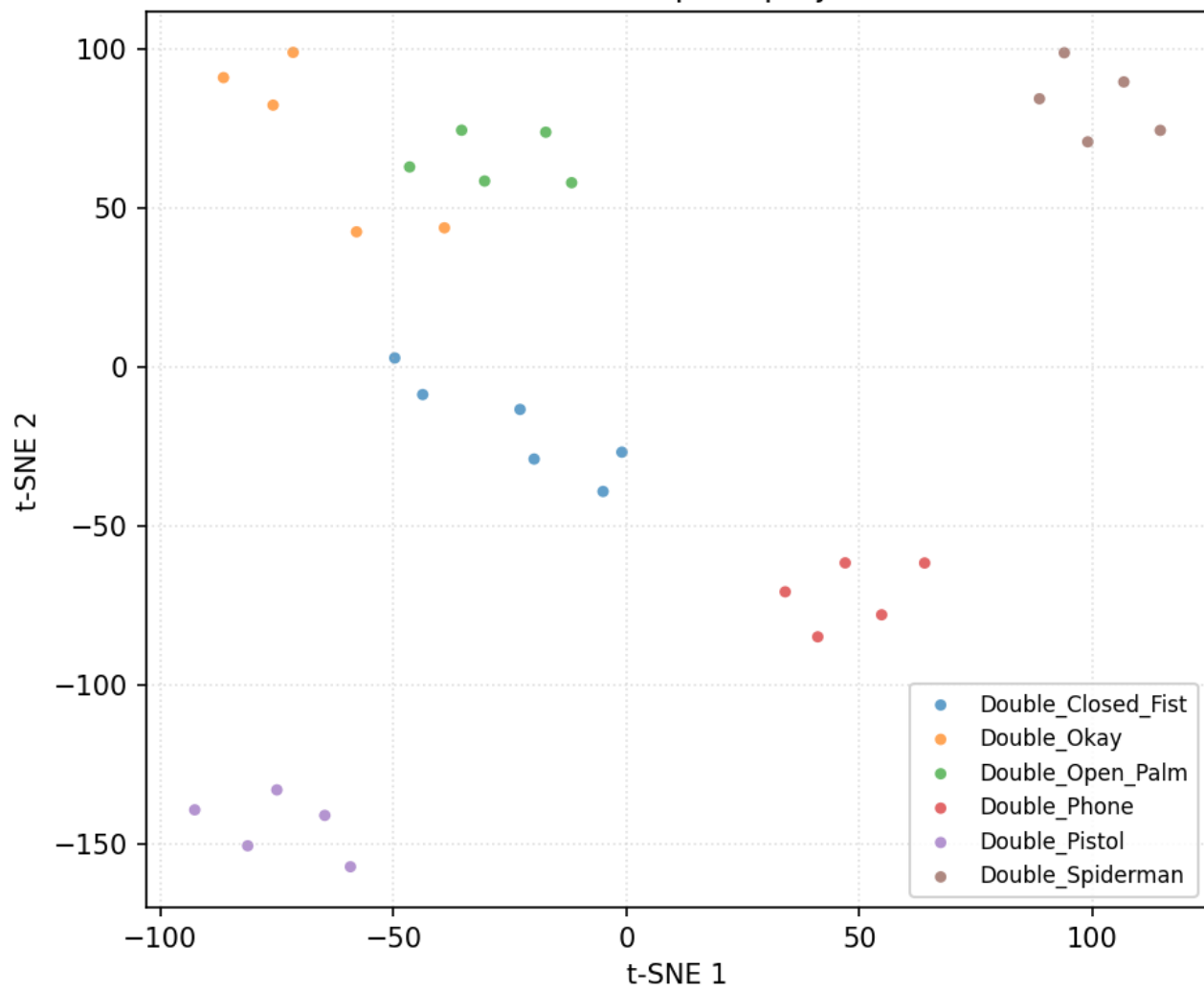


Data diagnostics — user 11_Mansh

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

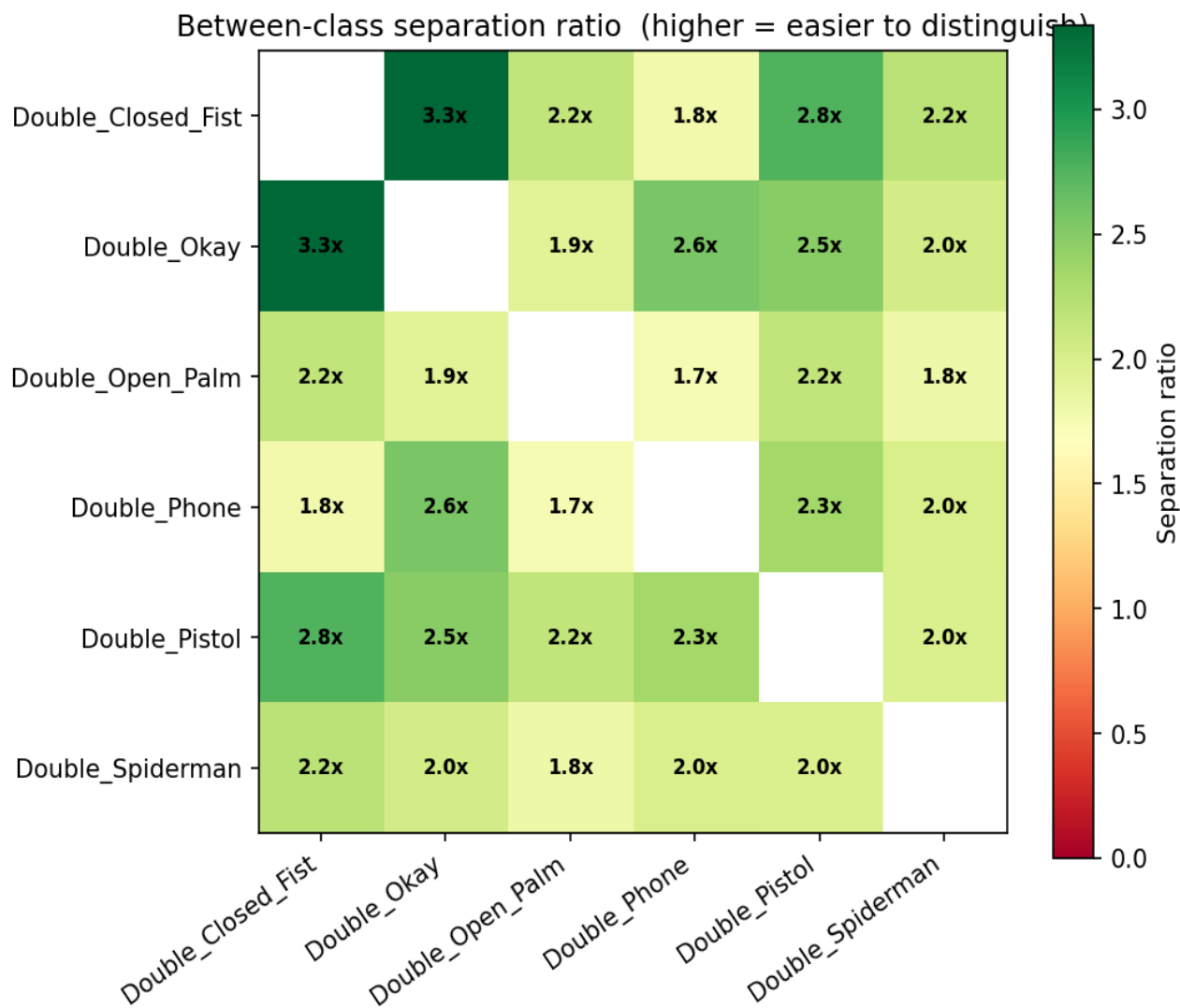
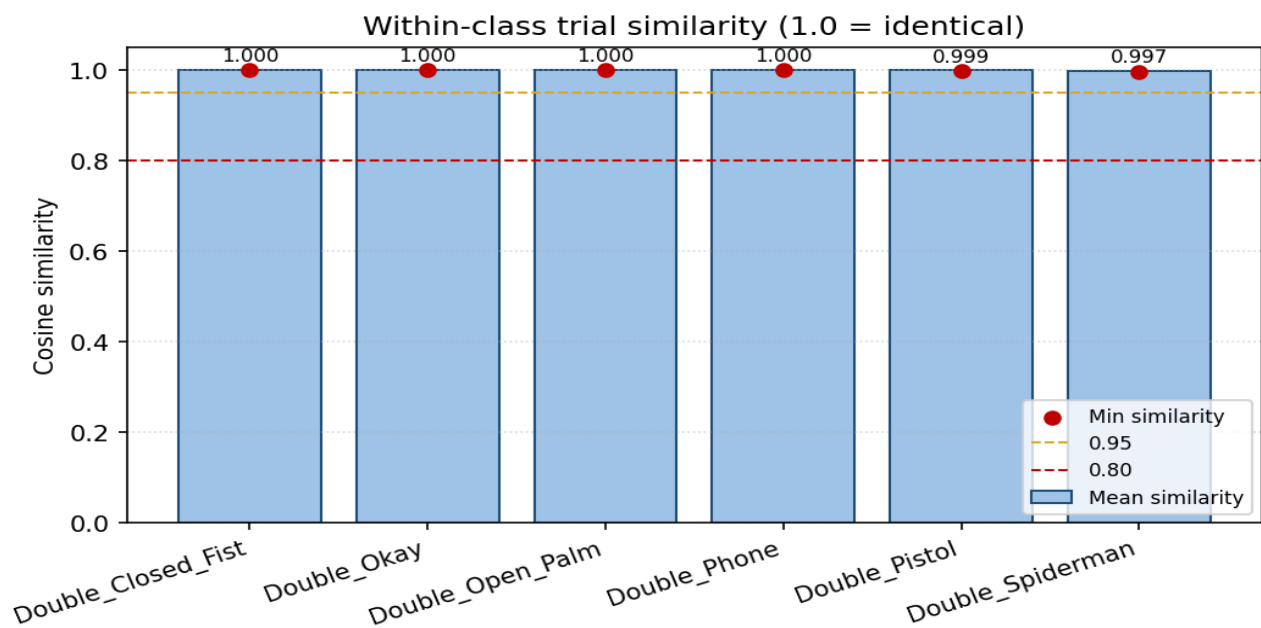


t-SNE feature space projection

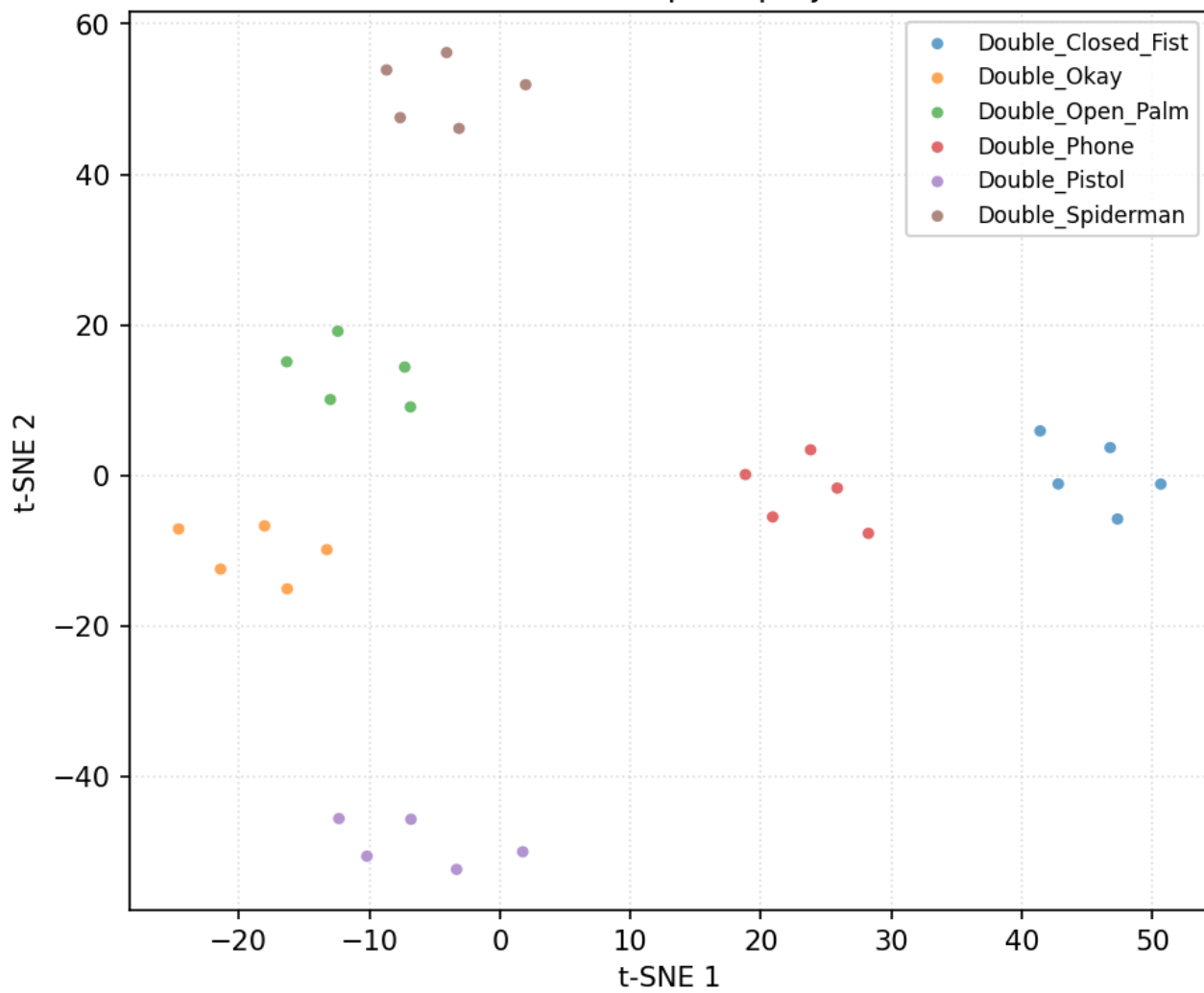


Data diagnostics — user 12_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

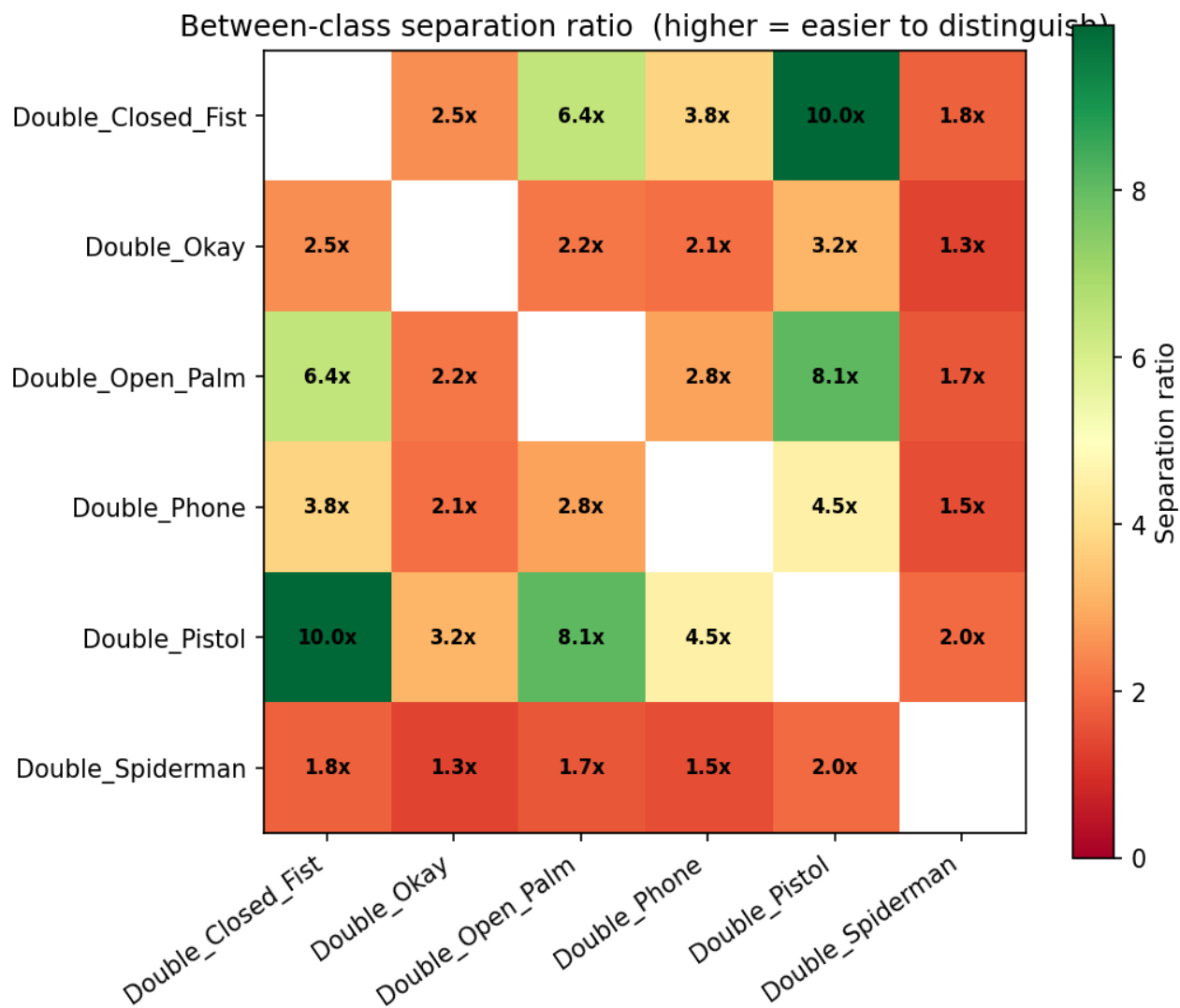
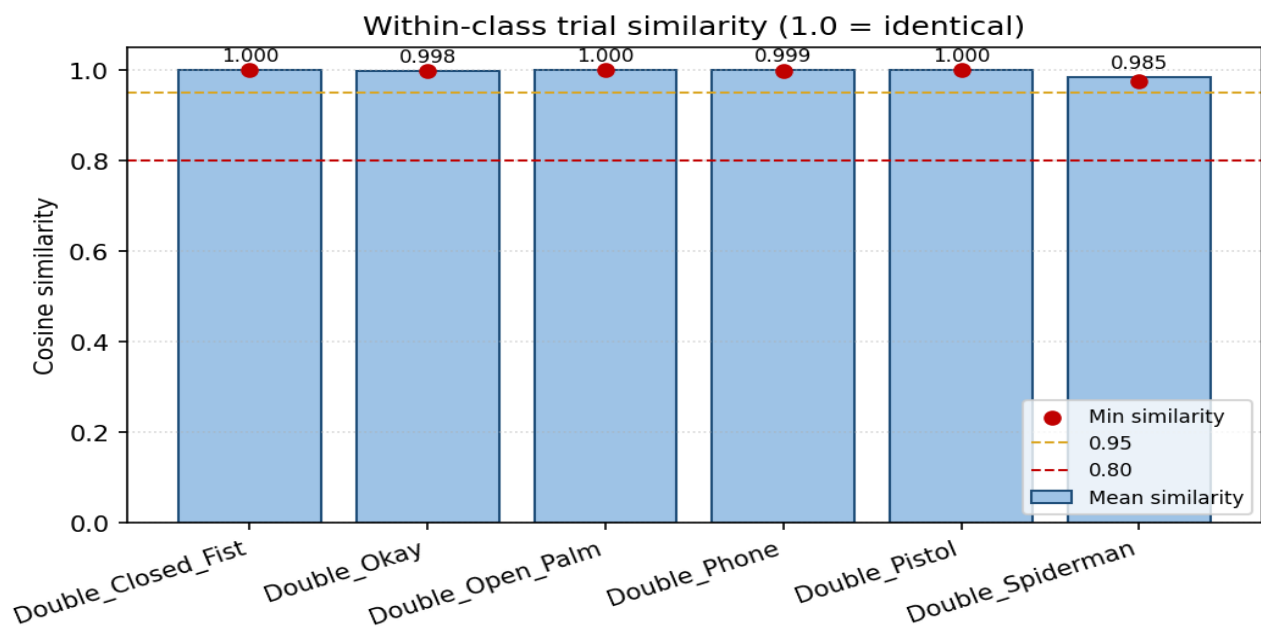


t-SNE feature space projection

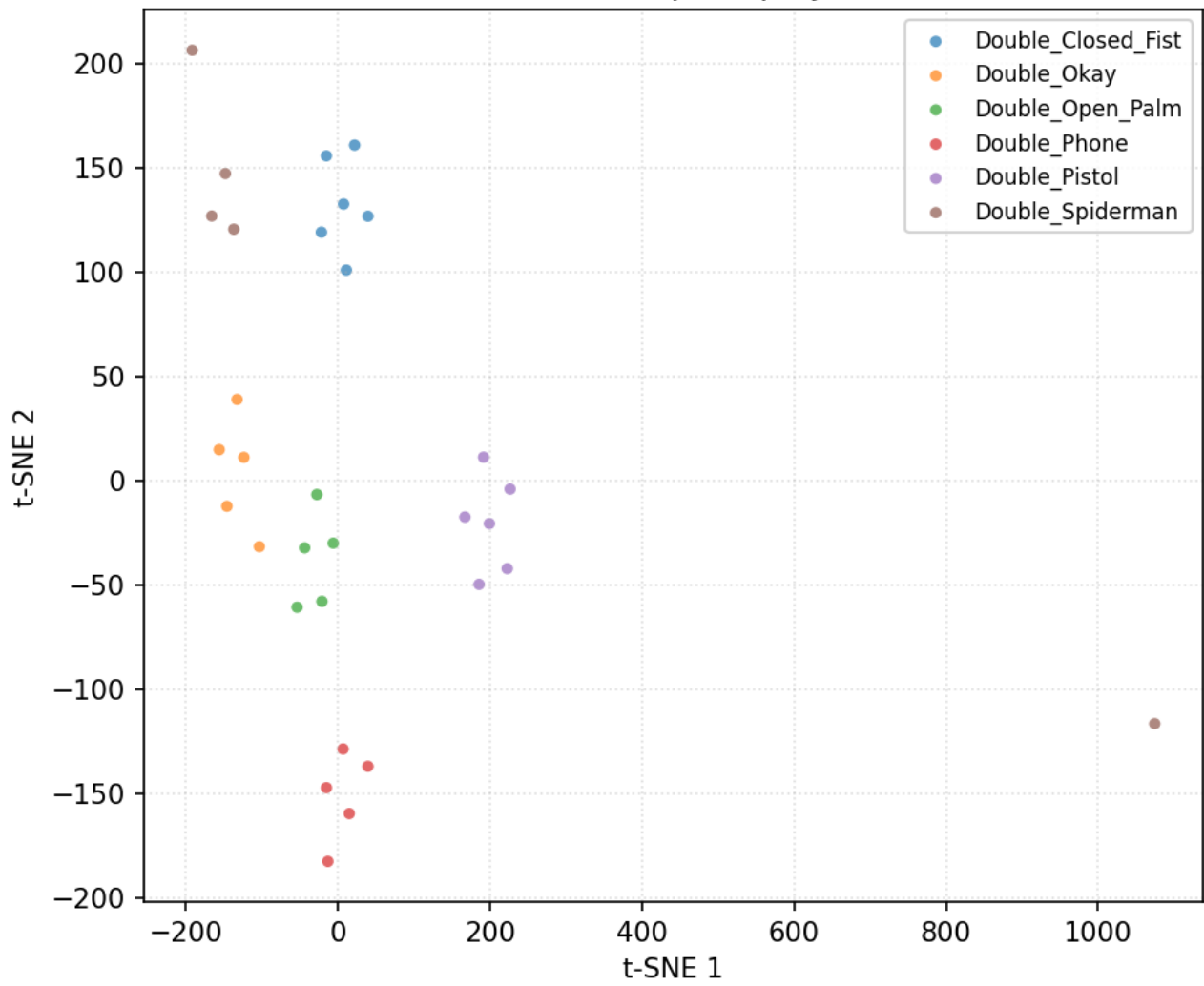


Data diagnostics — user 13_Nat

32 trials, 6 classes (filtered + resampled, before per-trial normalisation).

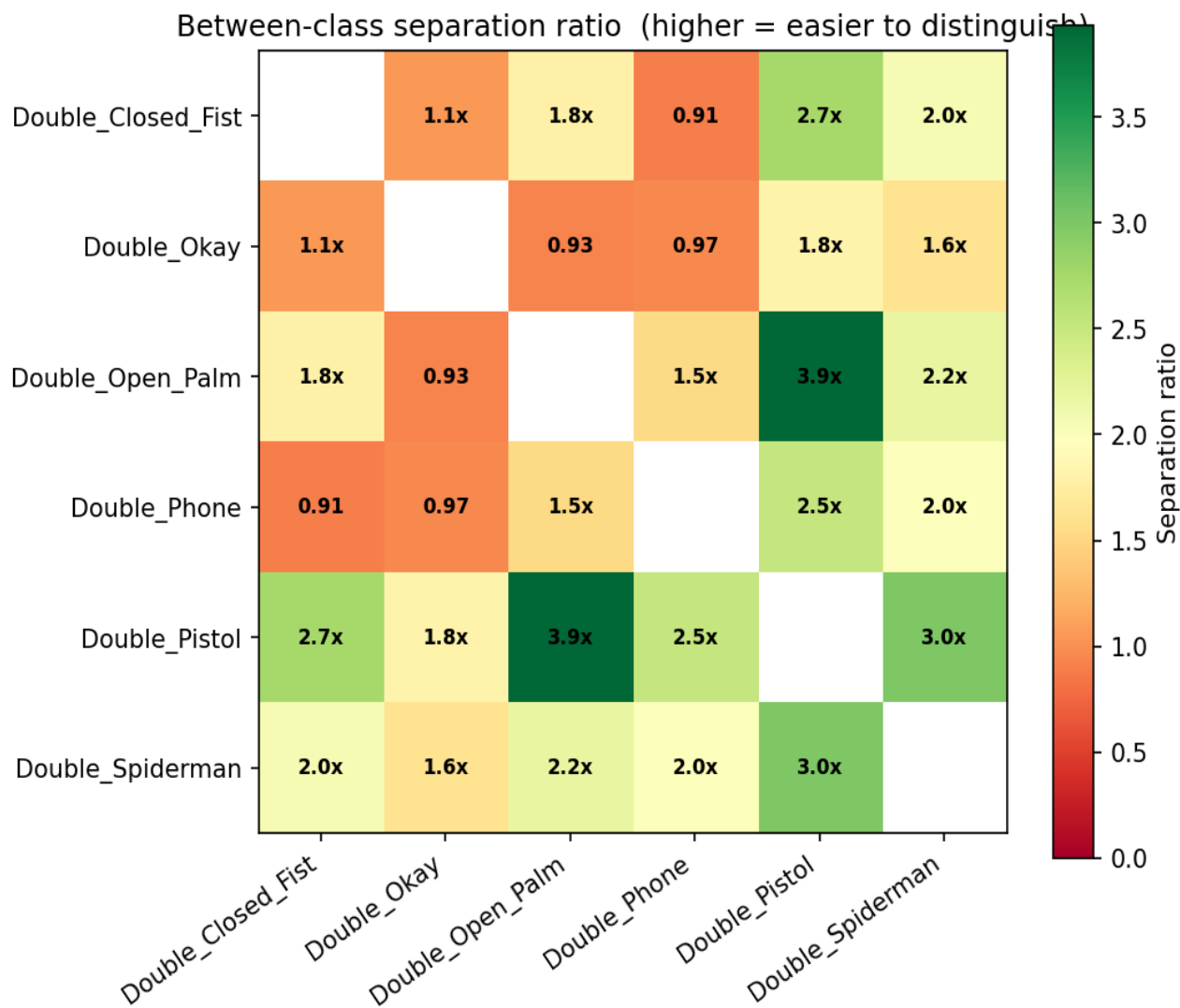
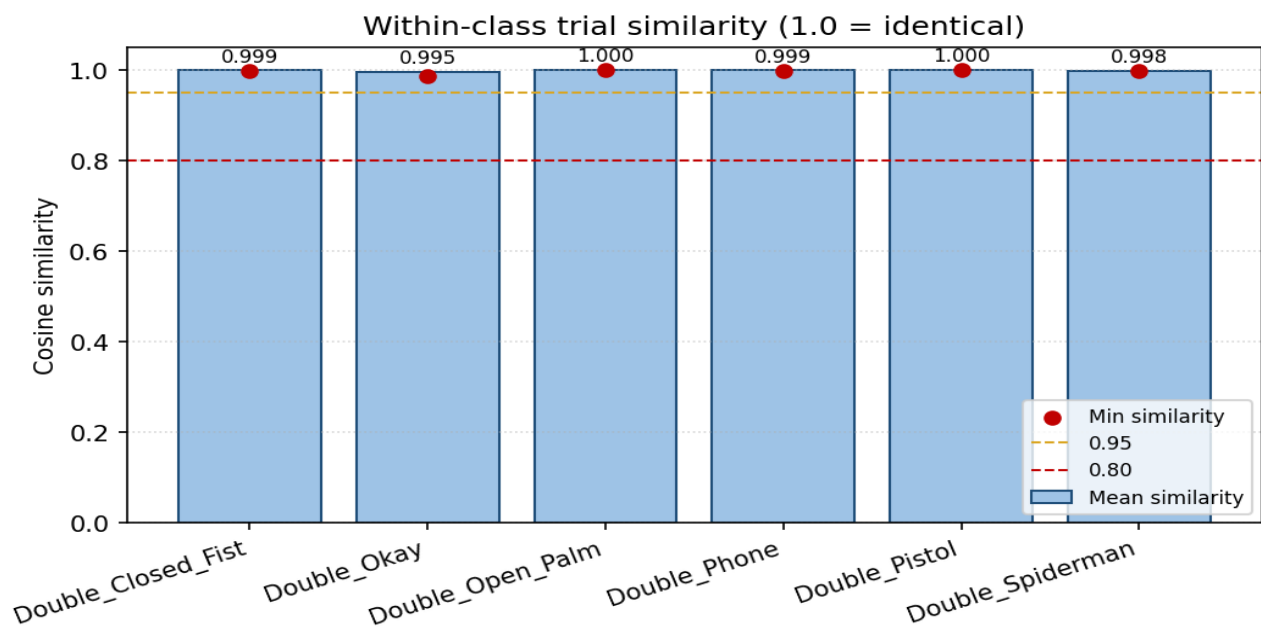


t-SNE feature space projection

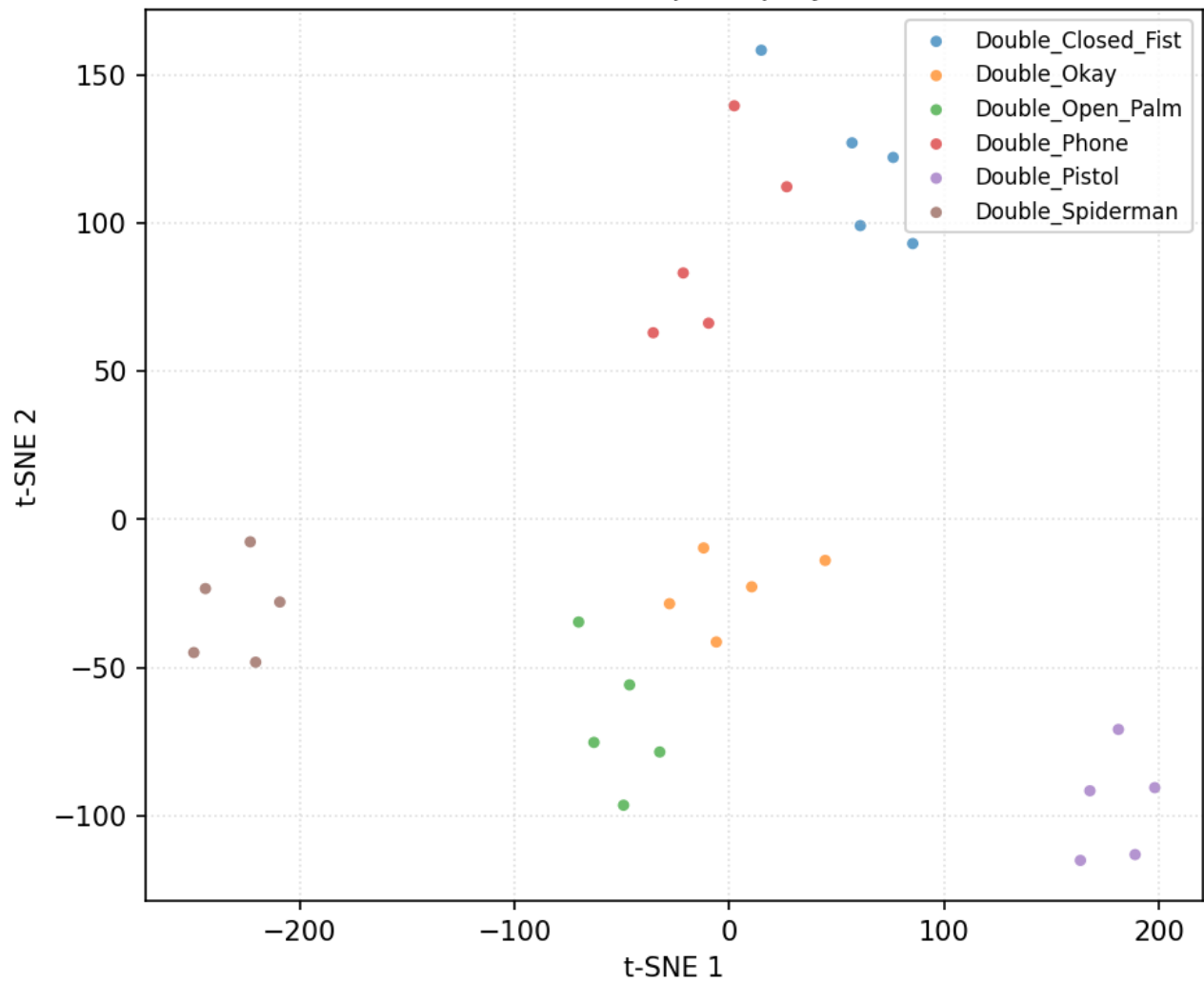


Data diagnostics — user 14_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

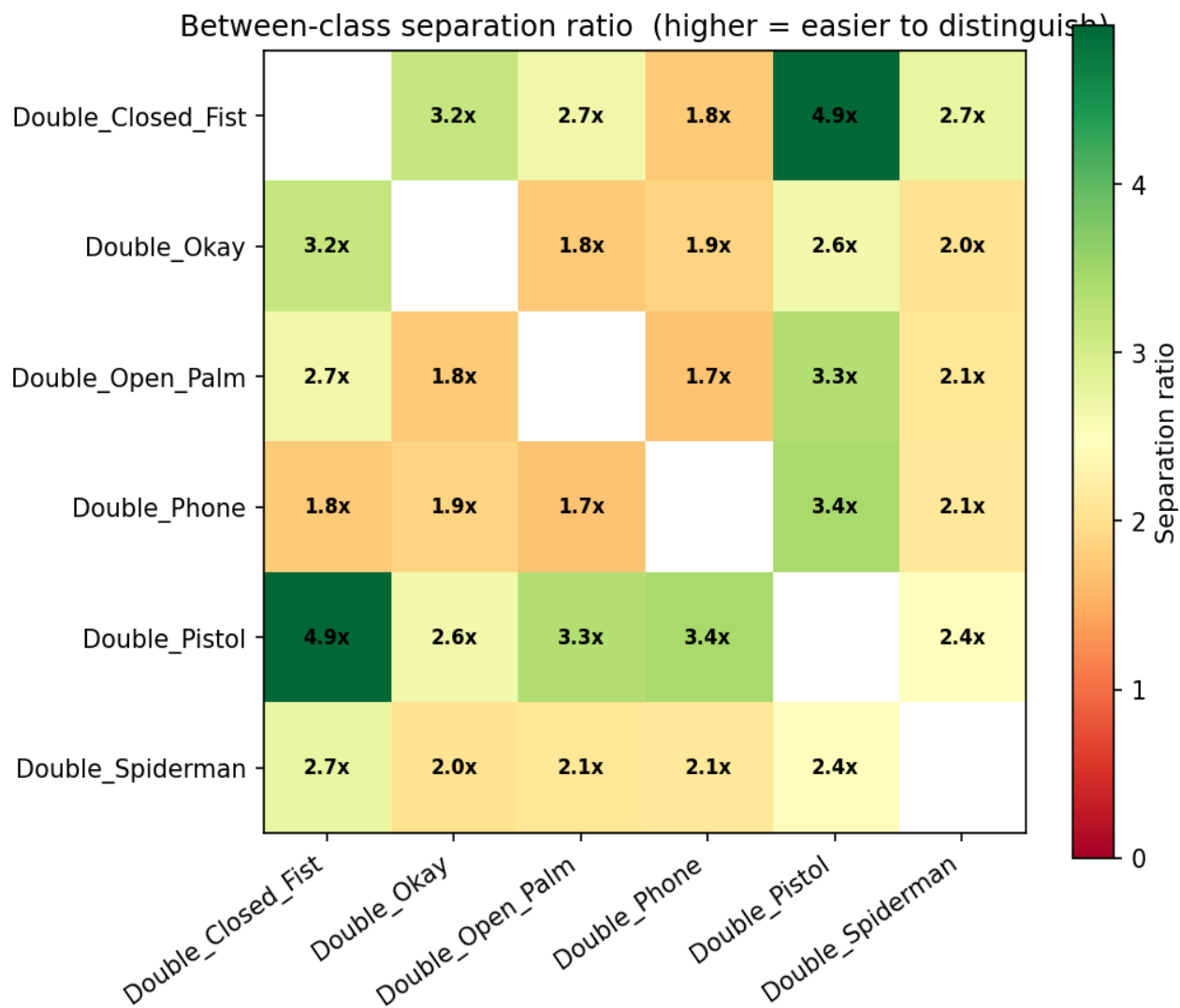
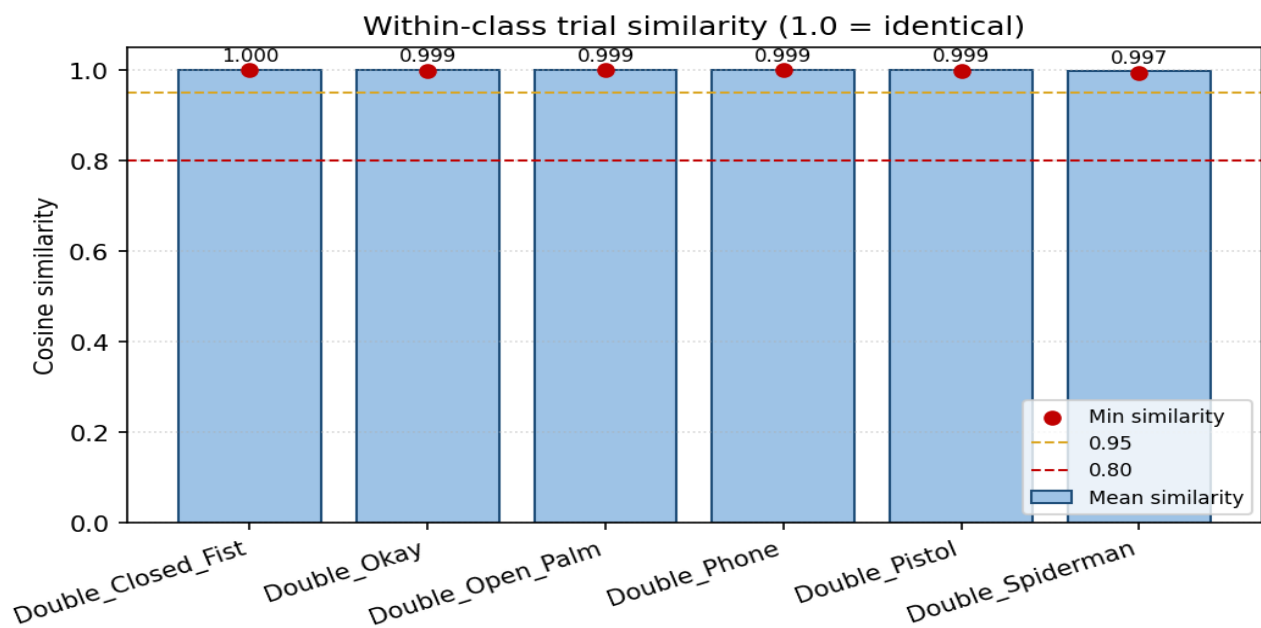


t-SNE feature space projection

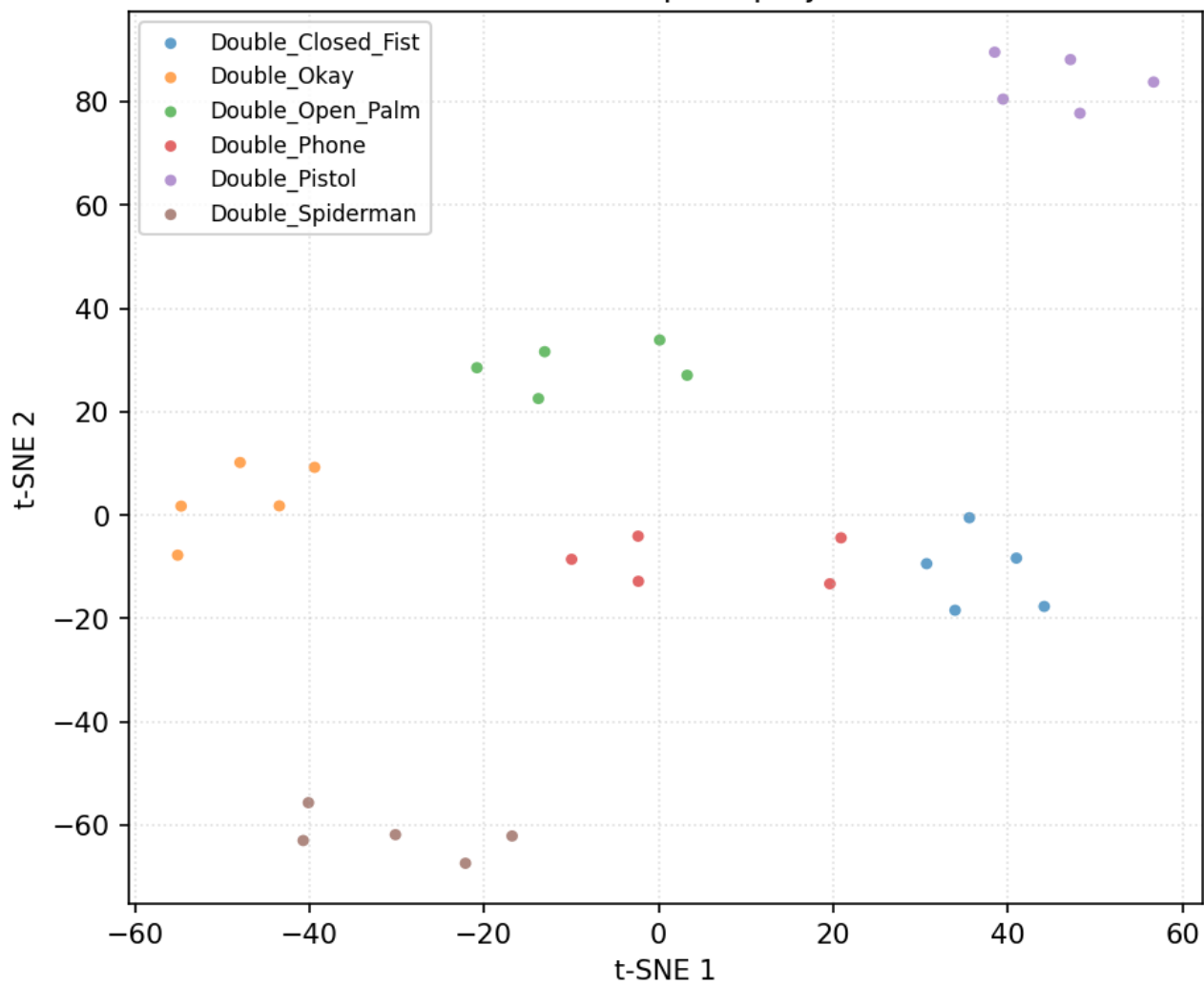


Data diagnostics — user 16_Bella

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

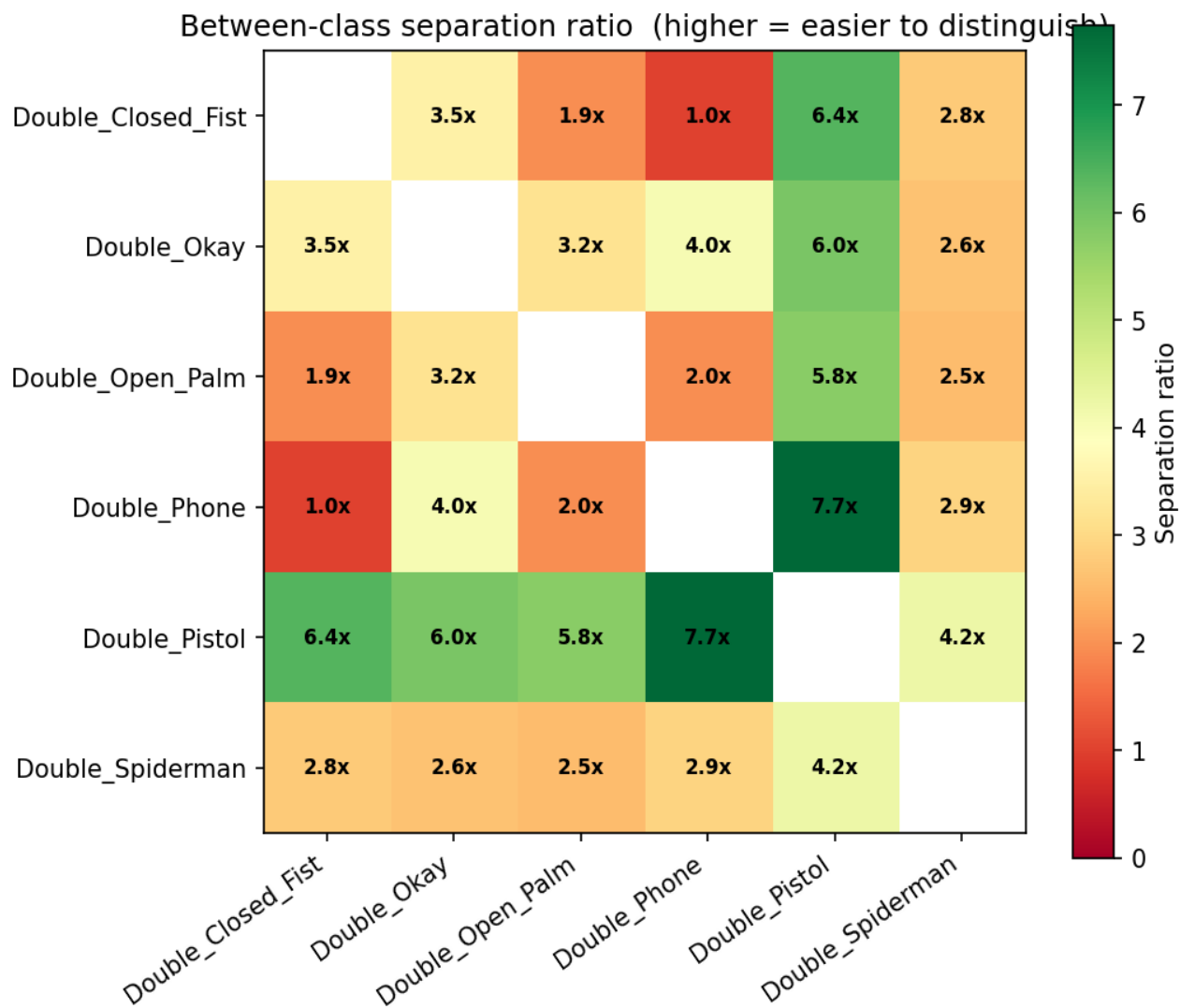
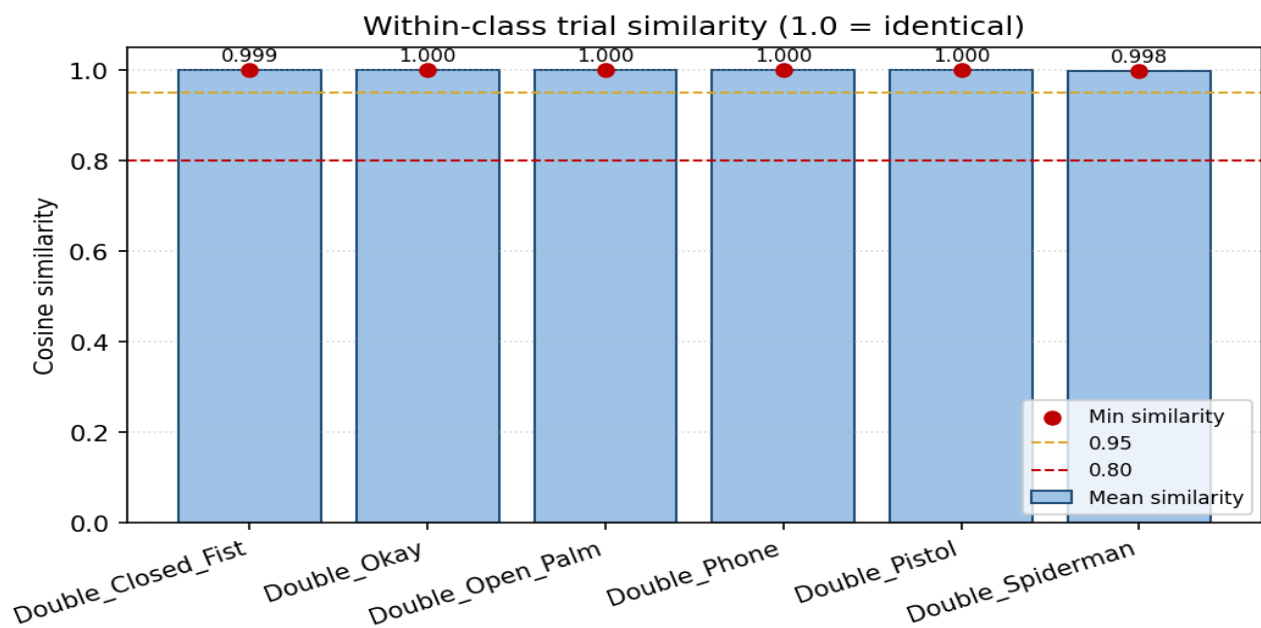


t-SNE feature space projection

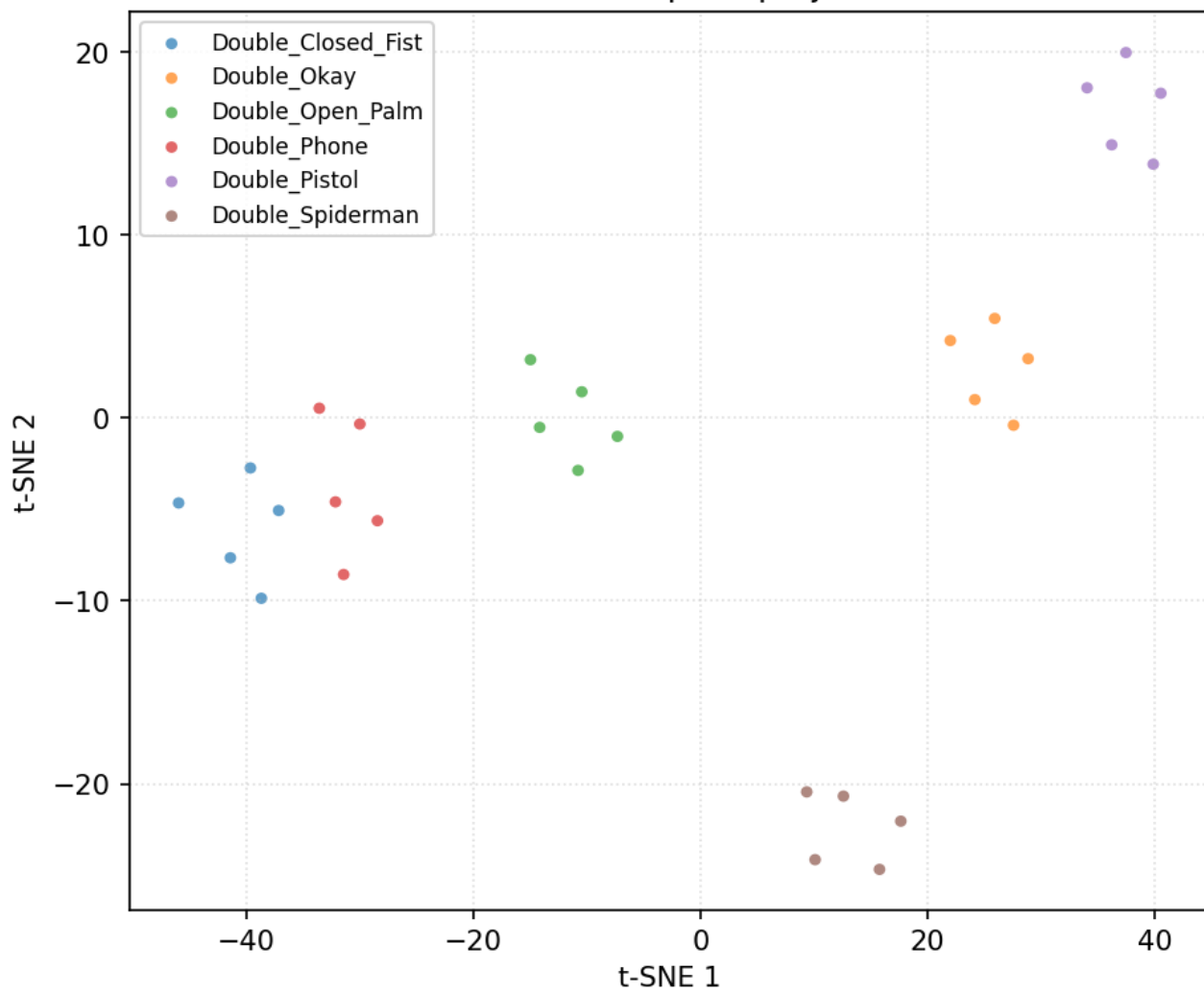


Data diagnostics — user 17_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

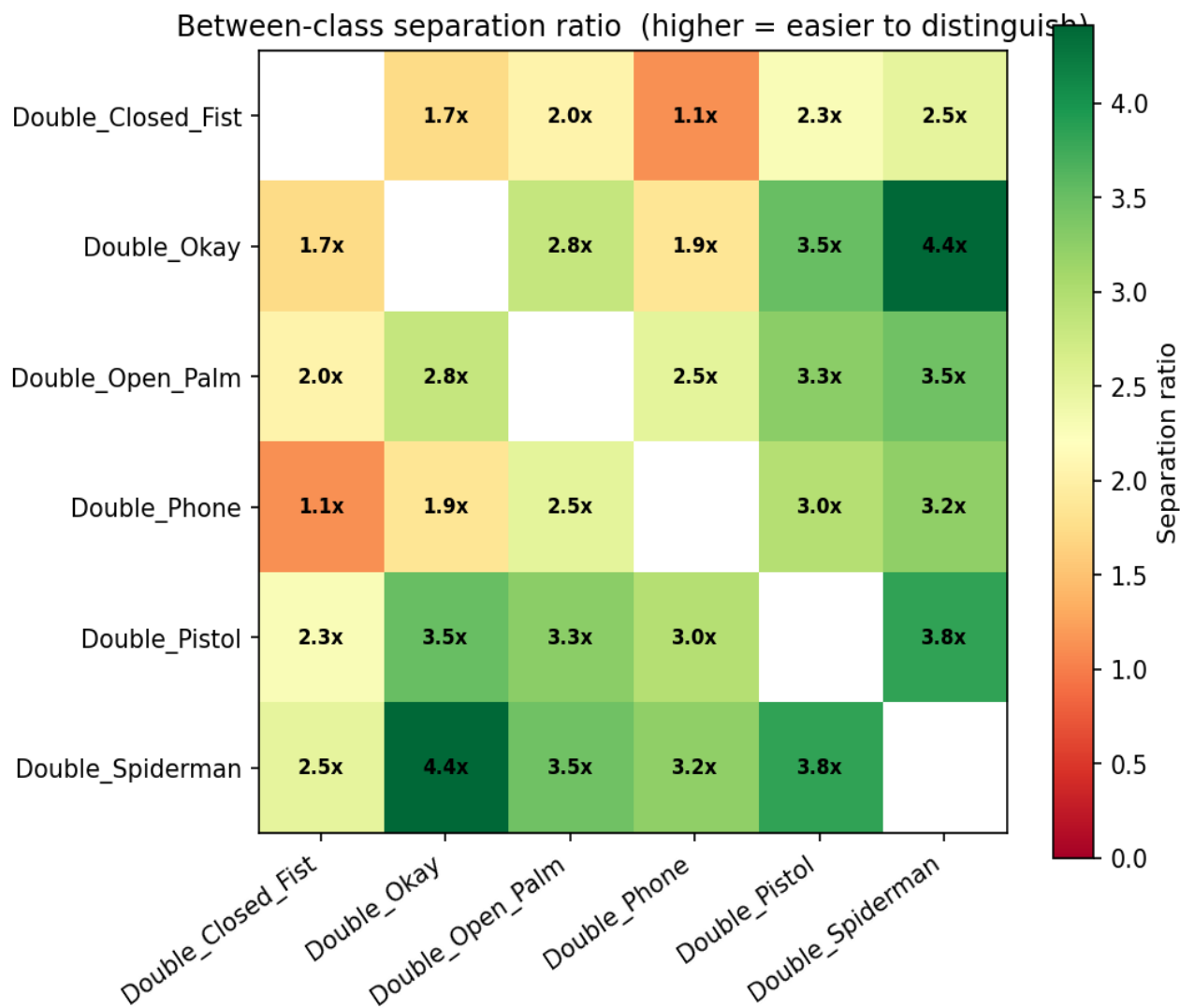
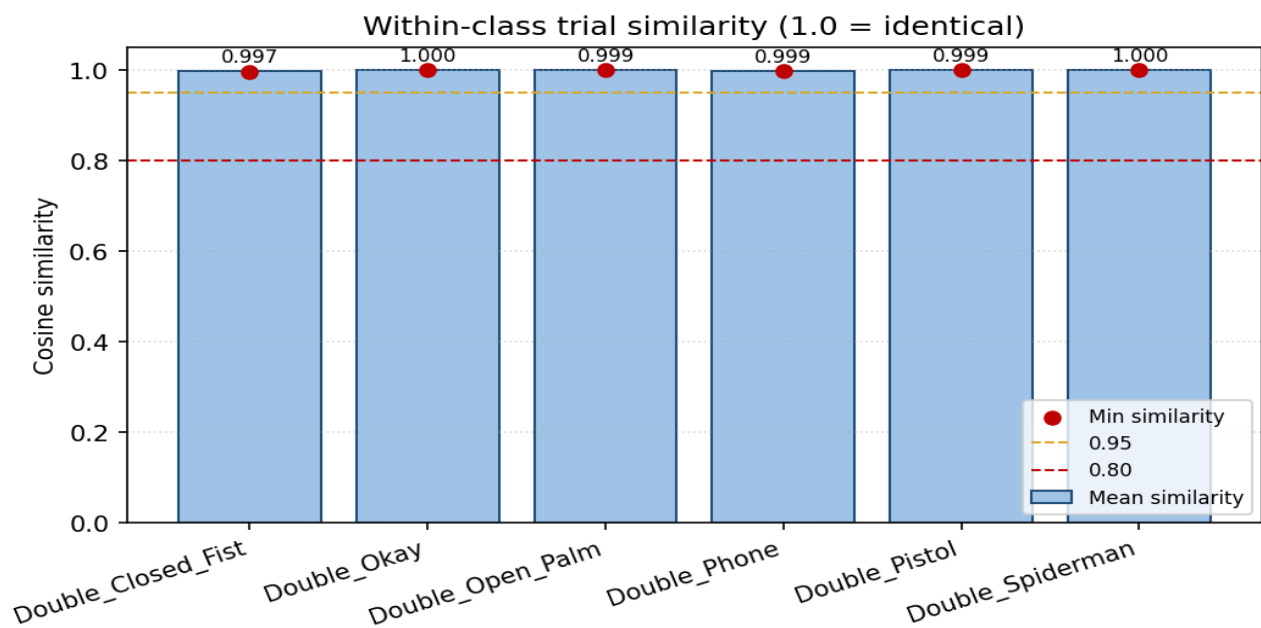


t-SNE feature space projection

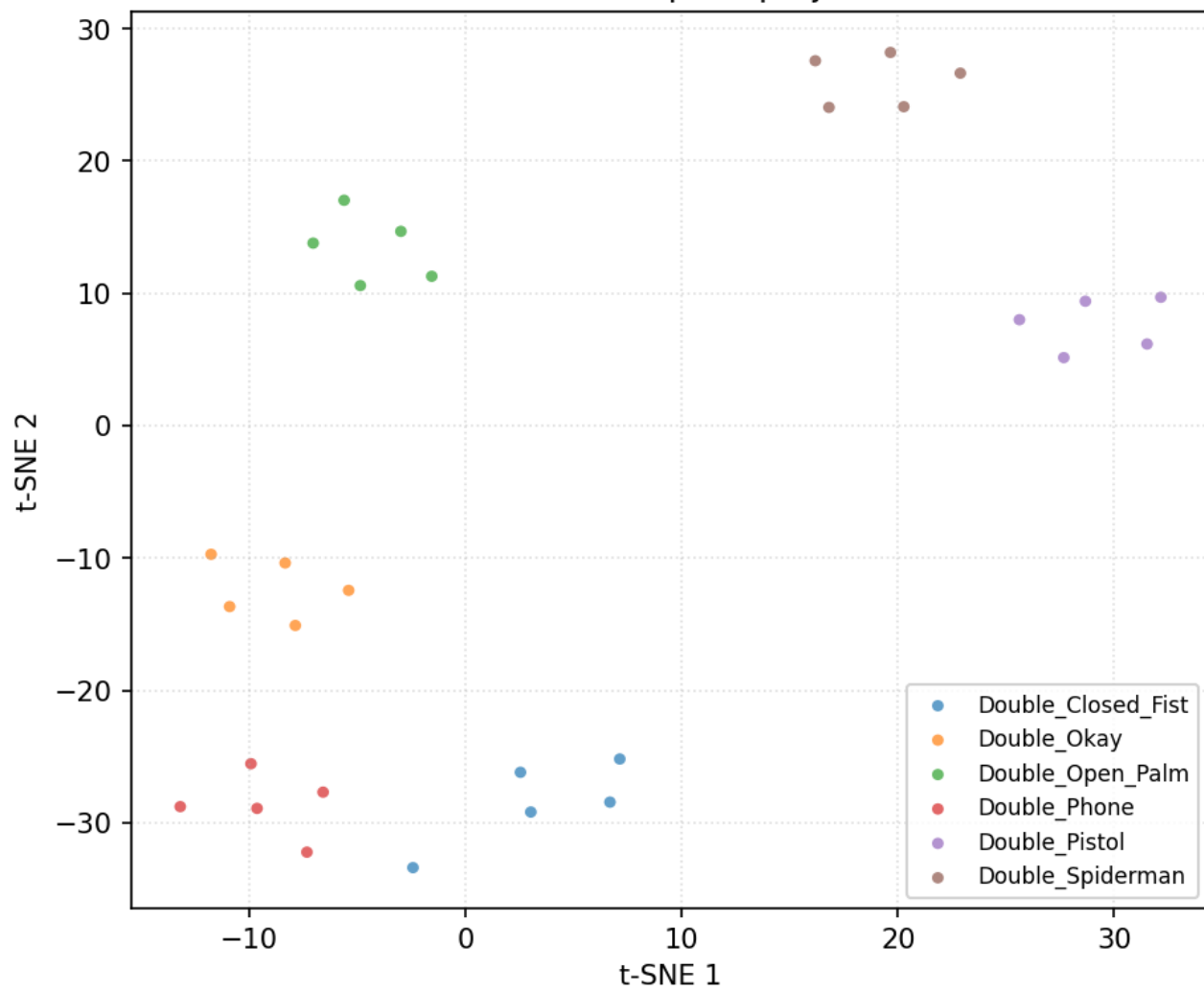


Data diagnostics — user 1_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

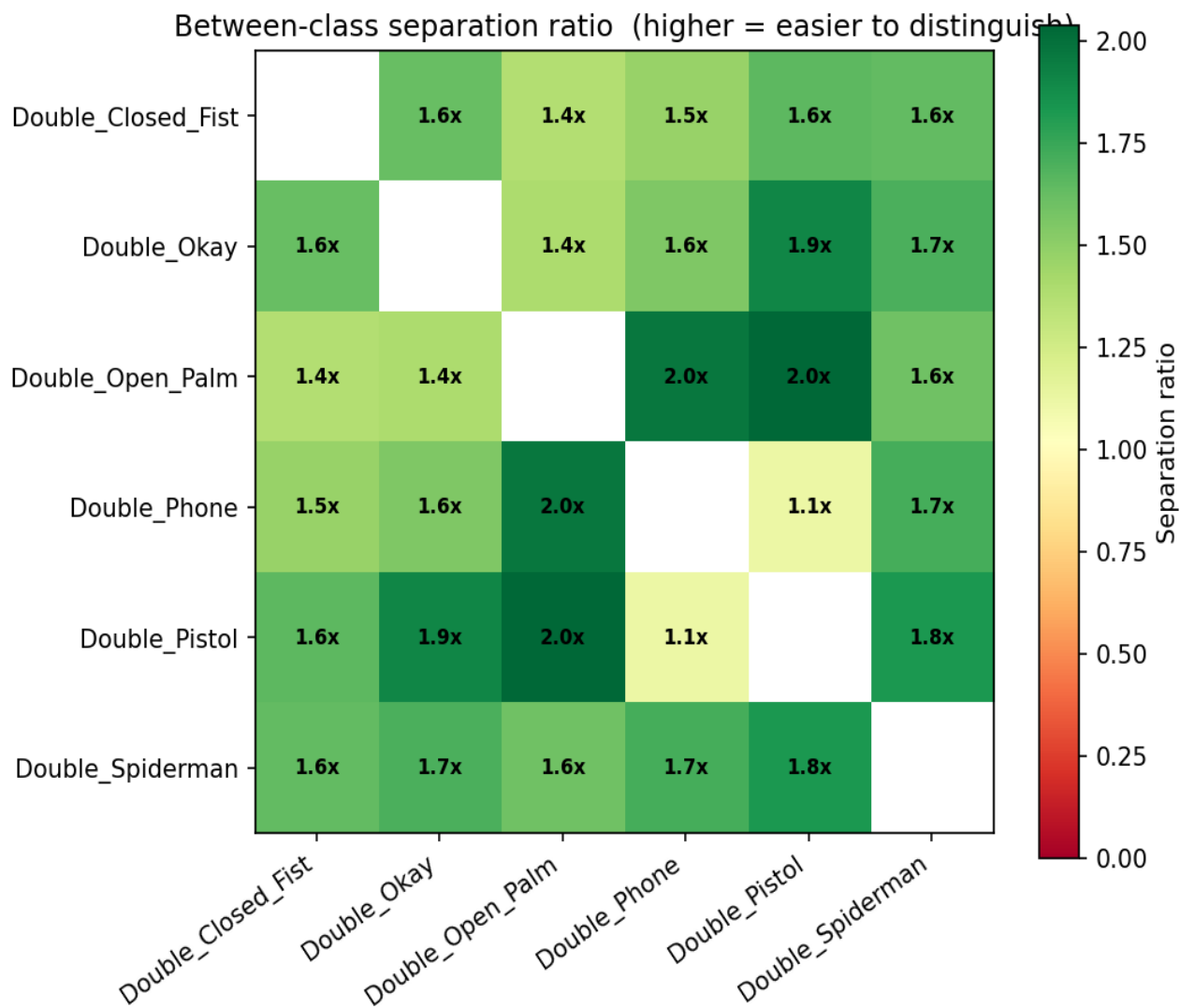
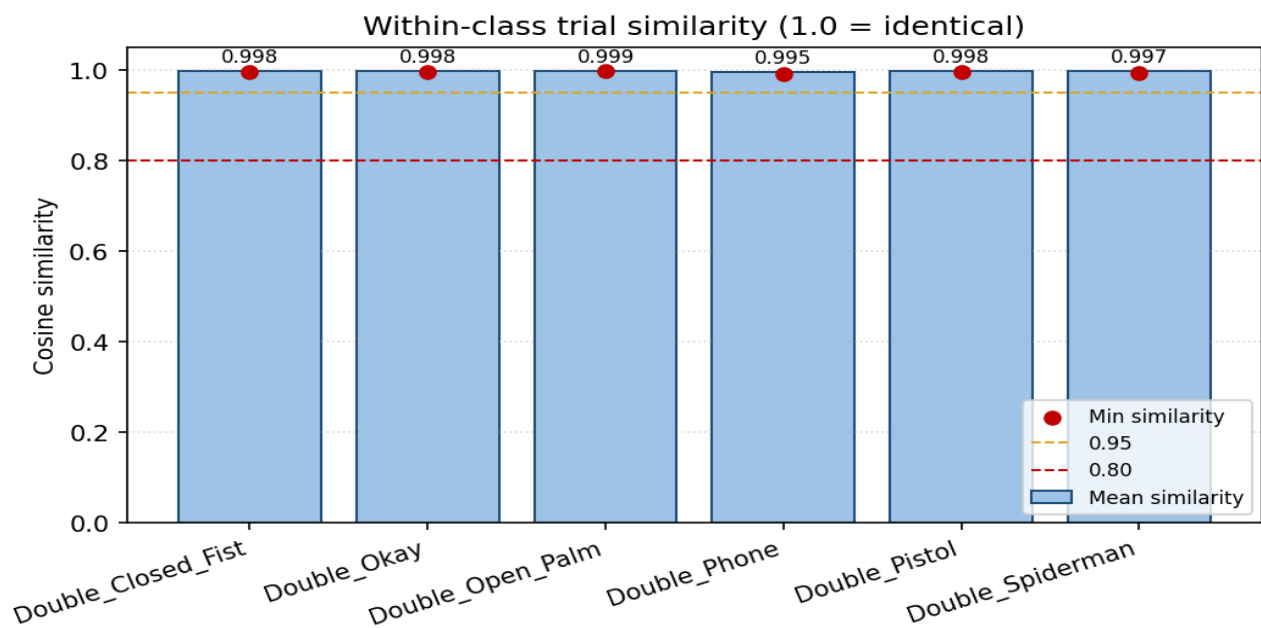


t-SNE feature space projection

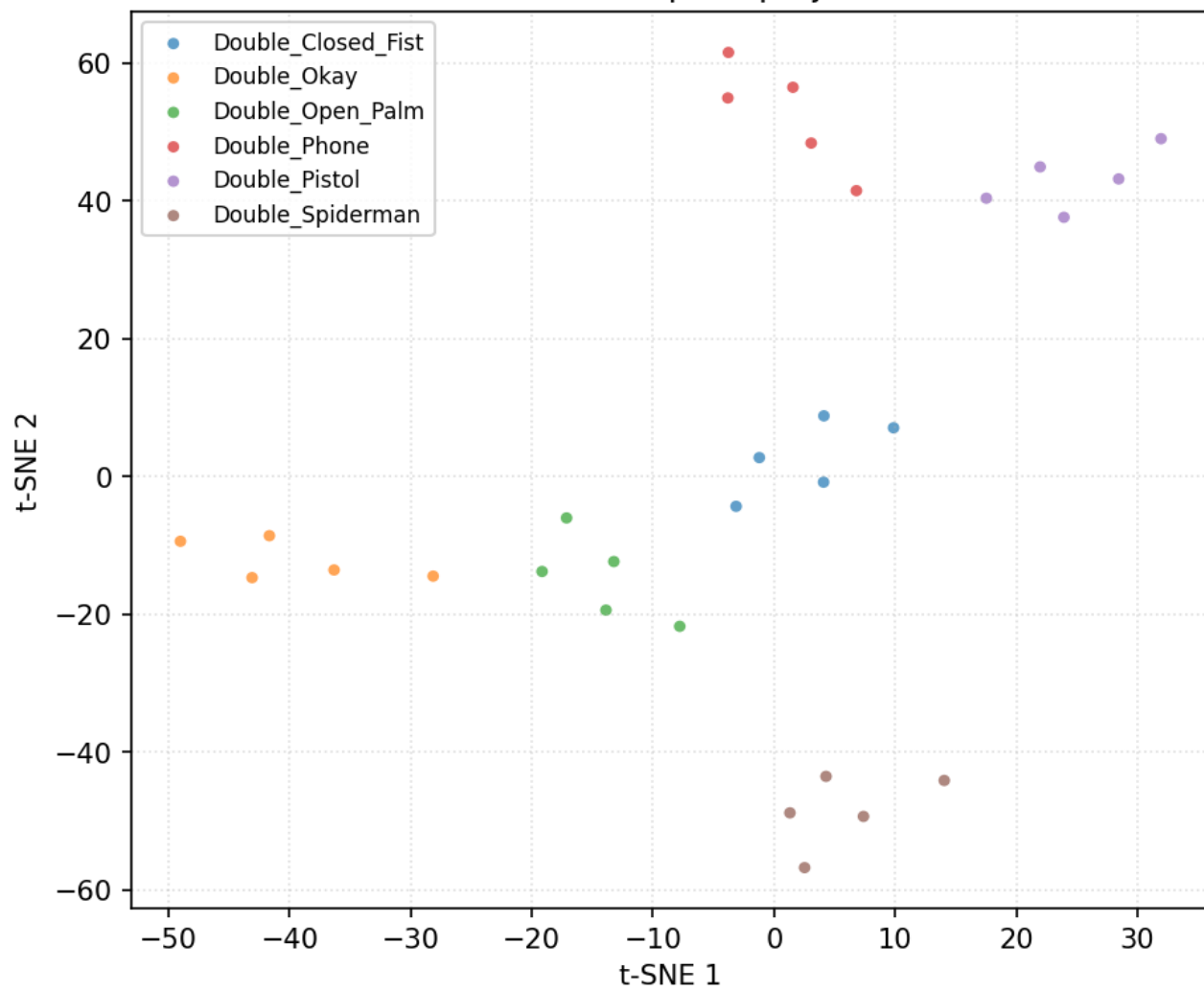


Data diagnostics — user 2_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

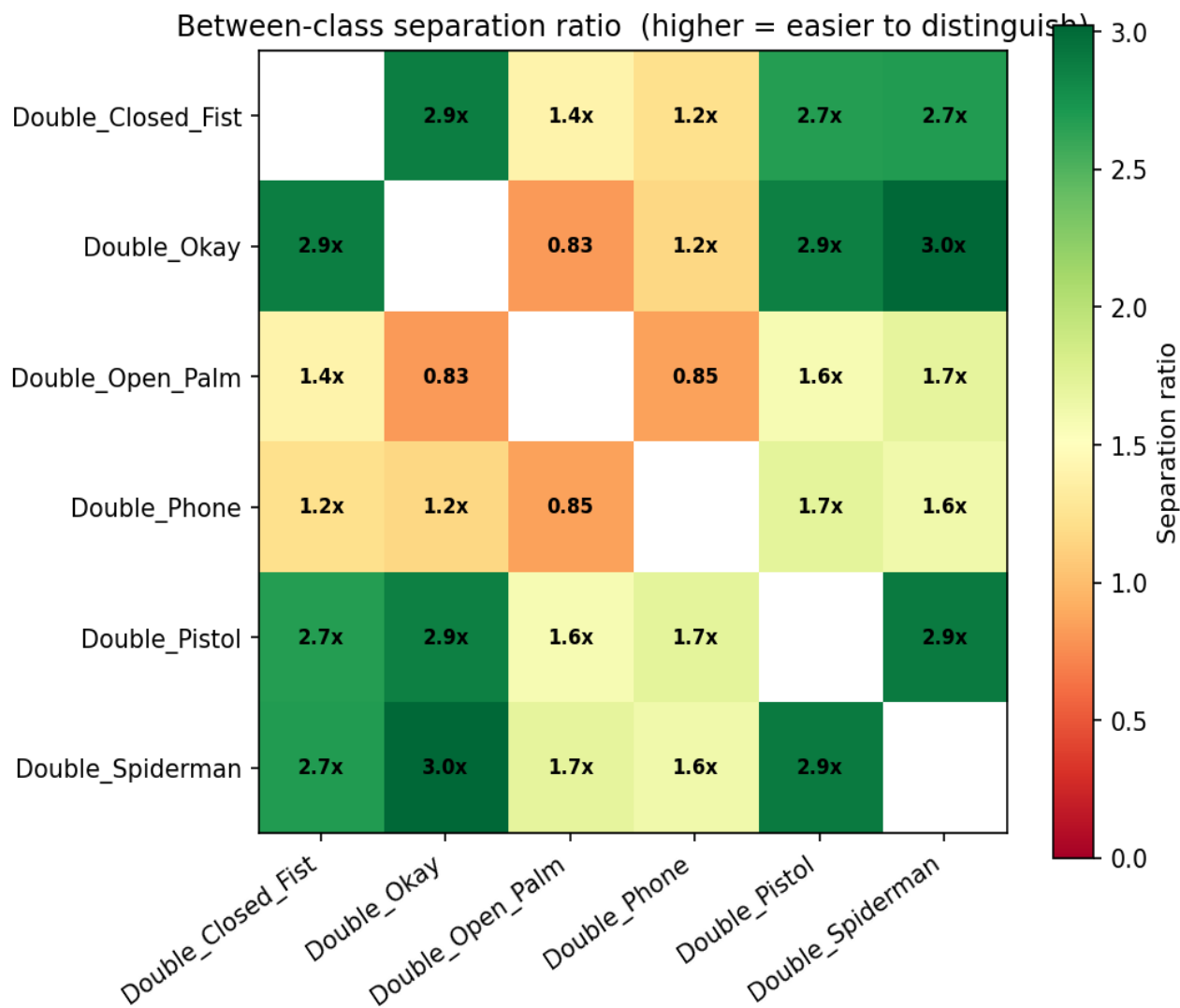
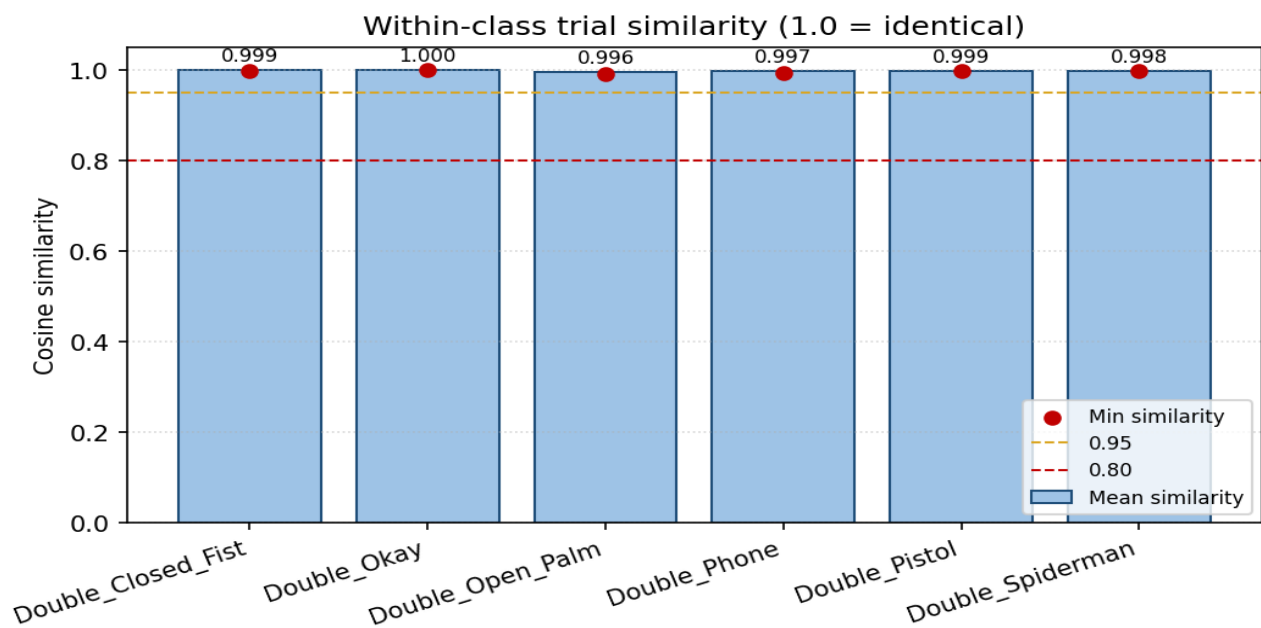


t-SNE feature space projection

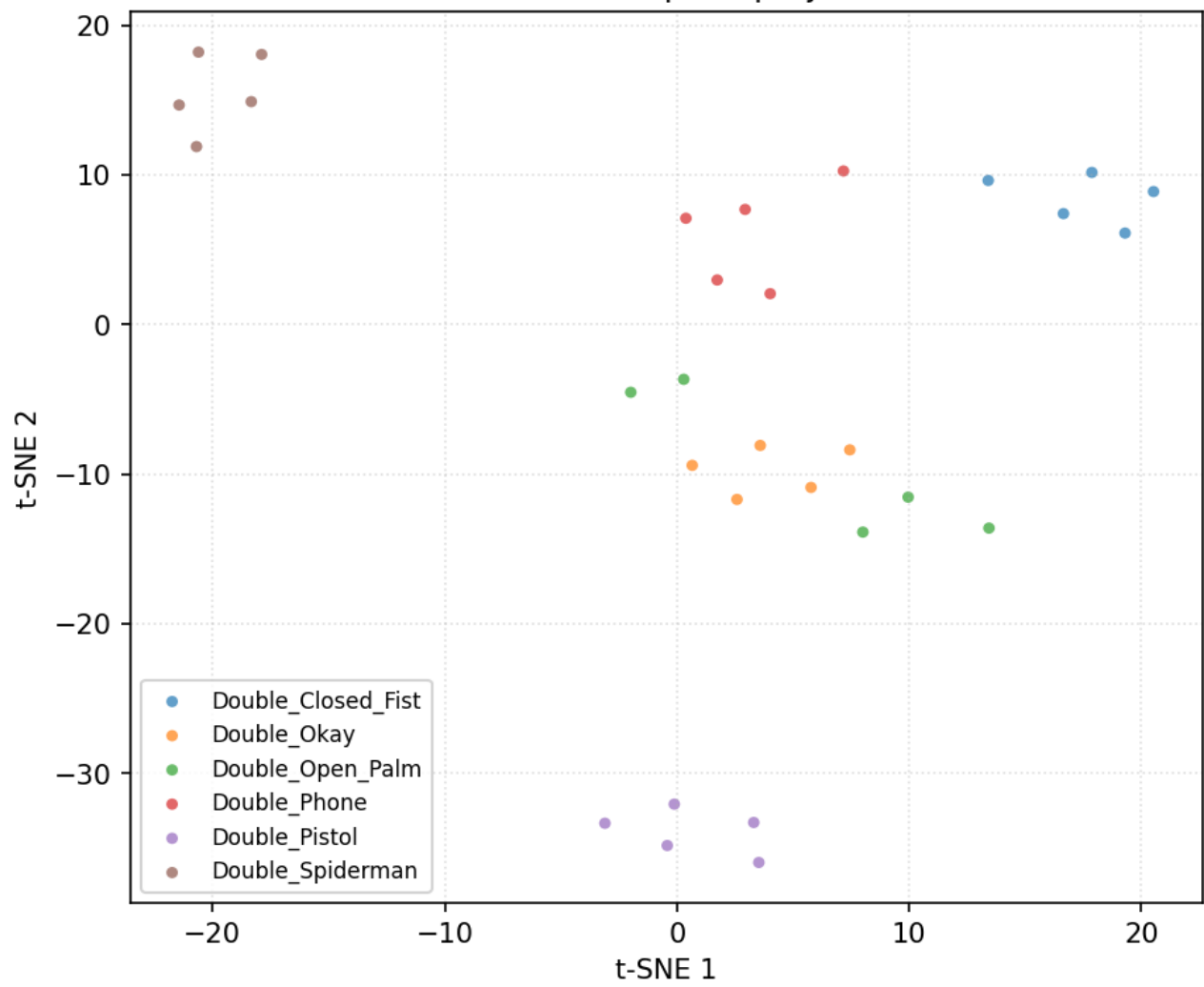


Data diagnostics — user 3_Anghad

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

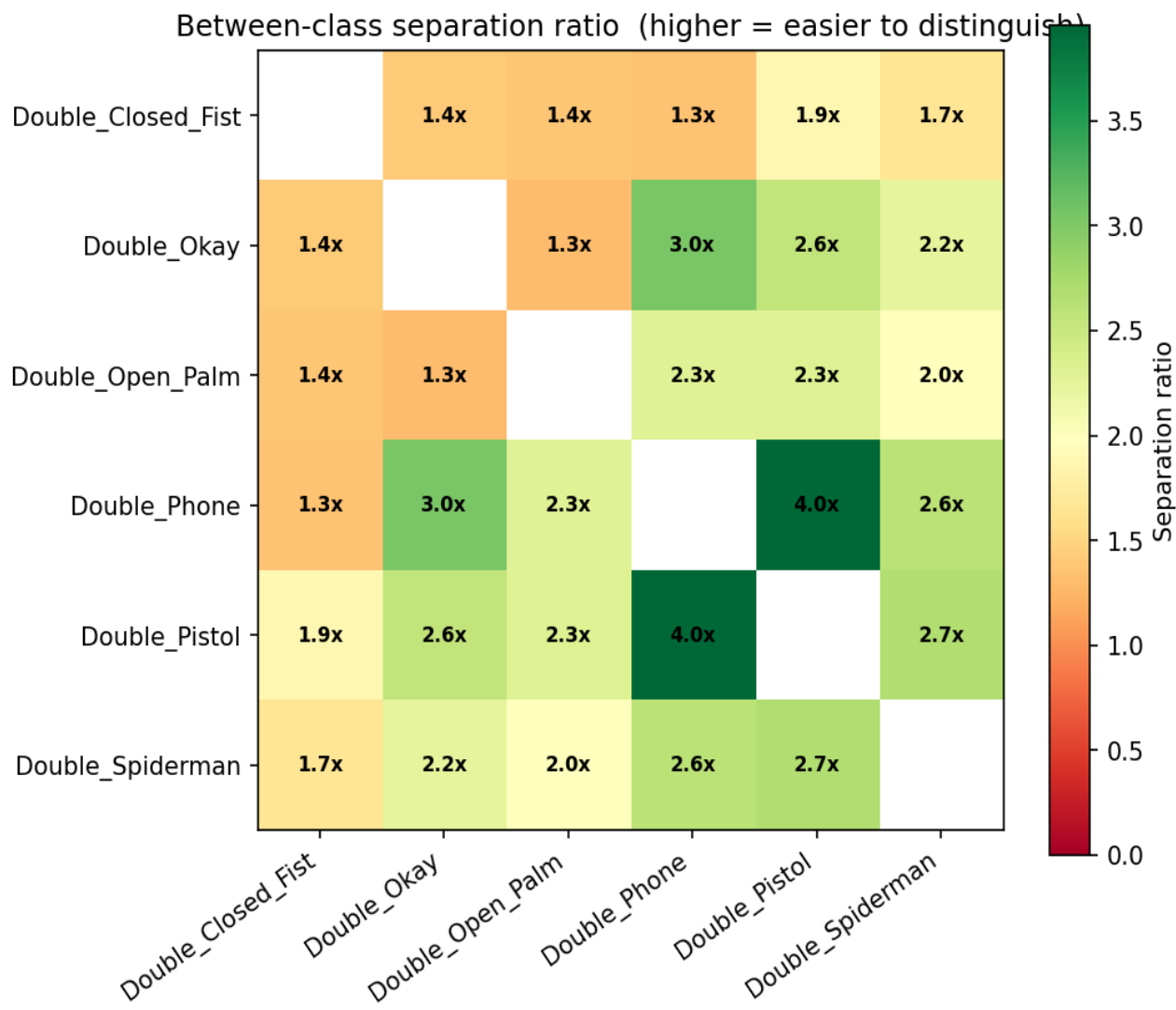
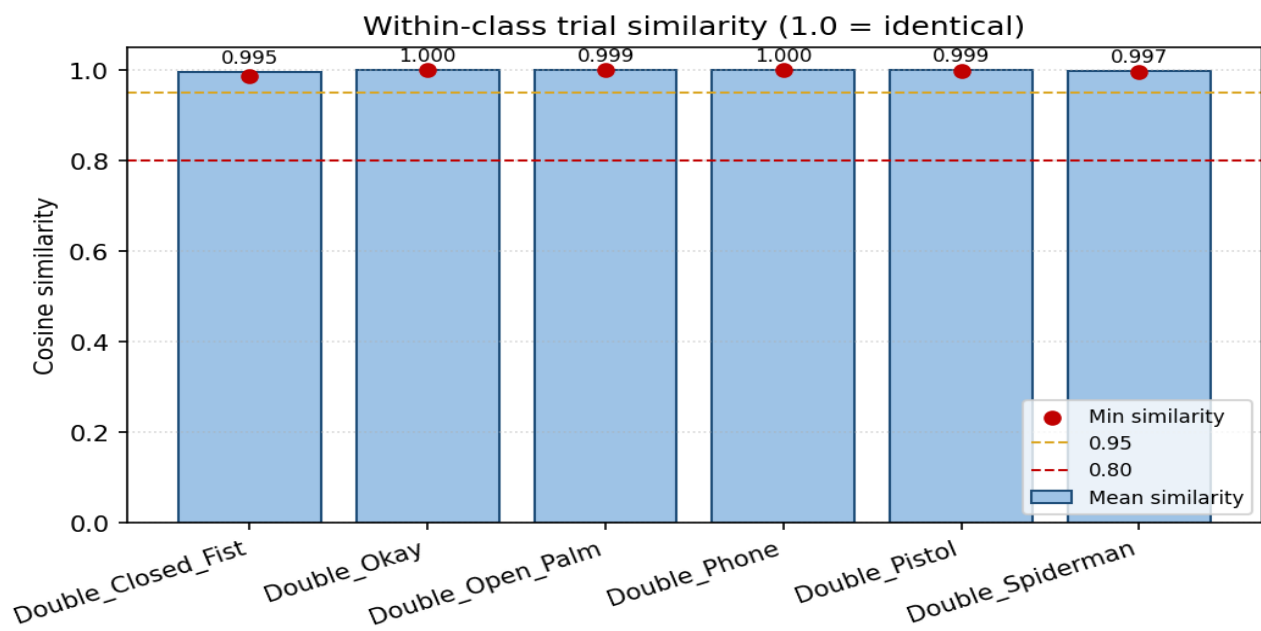


t-SNE feature space projection

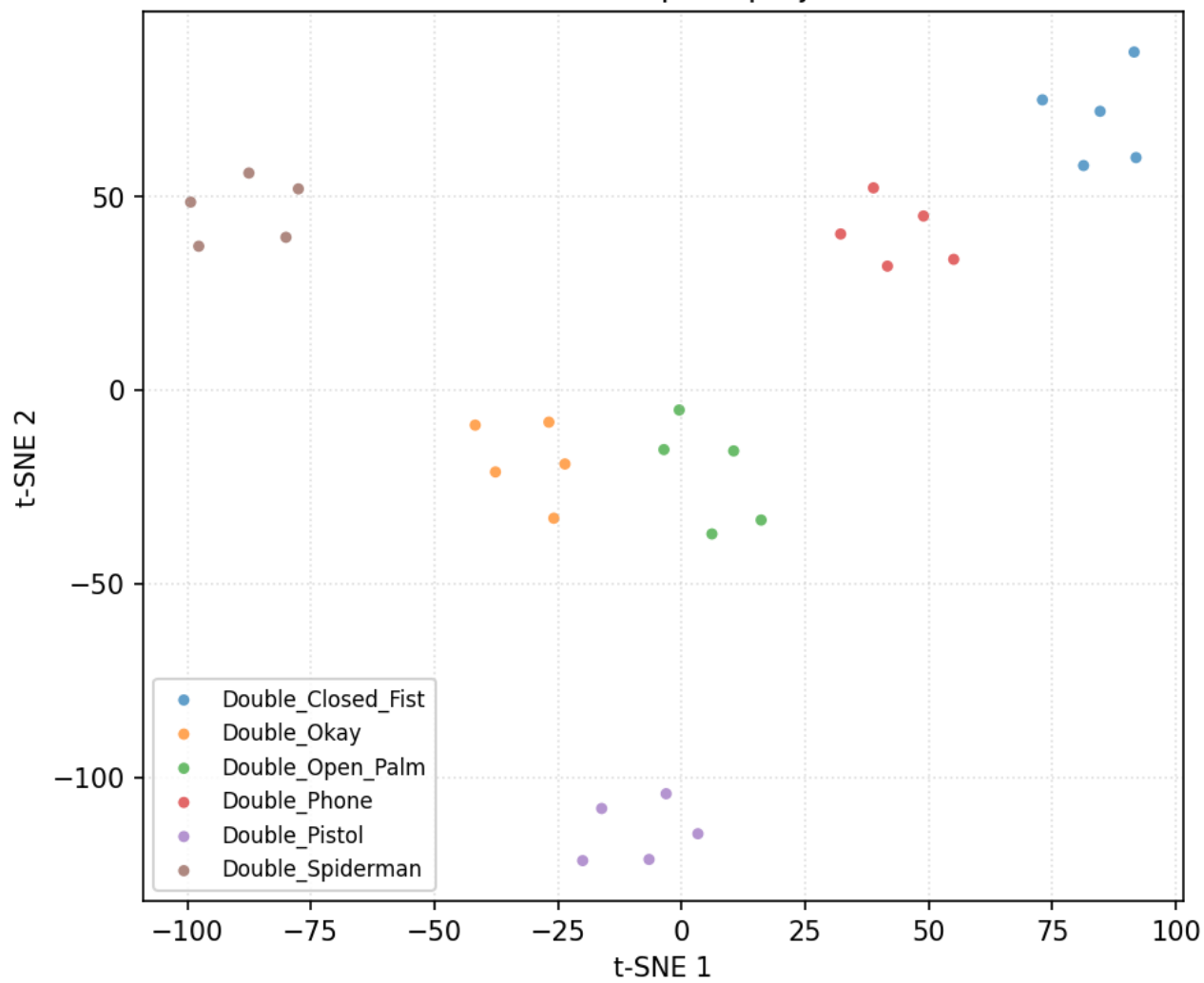


Data diagnostics — user 4_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

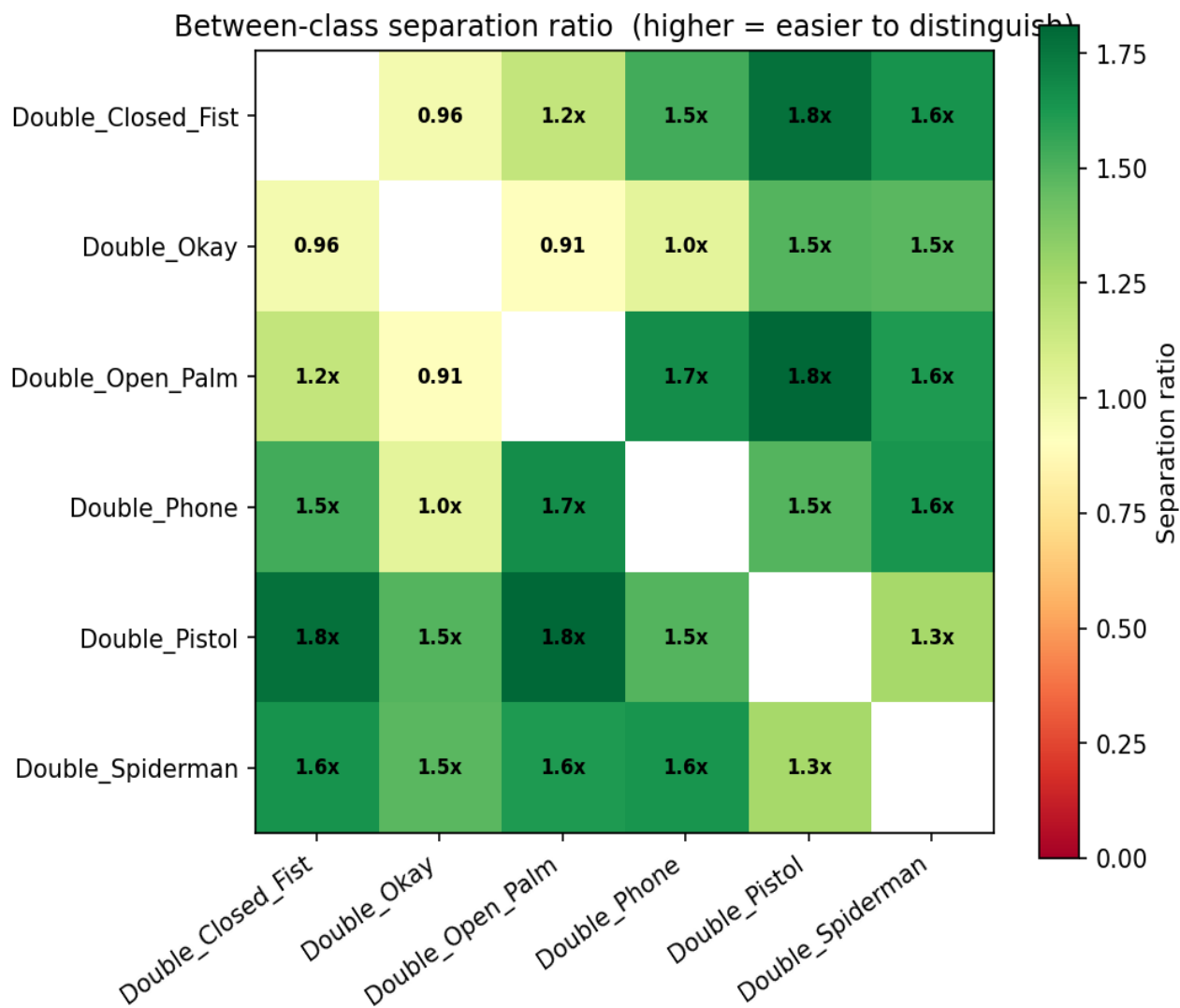
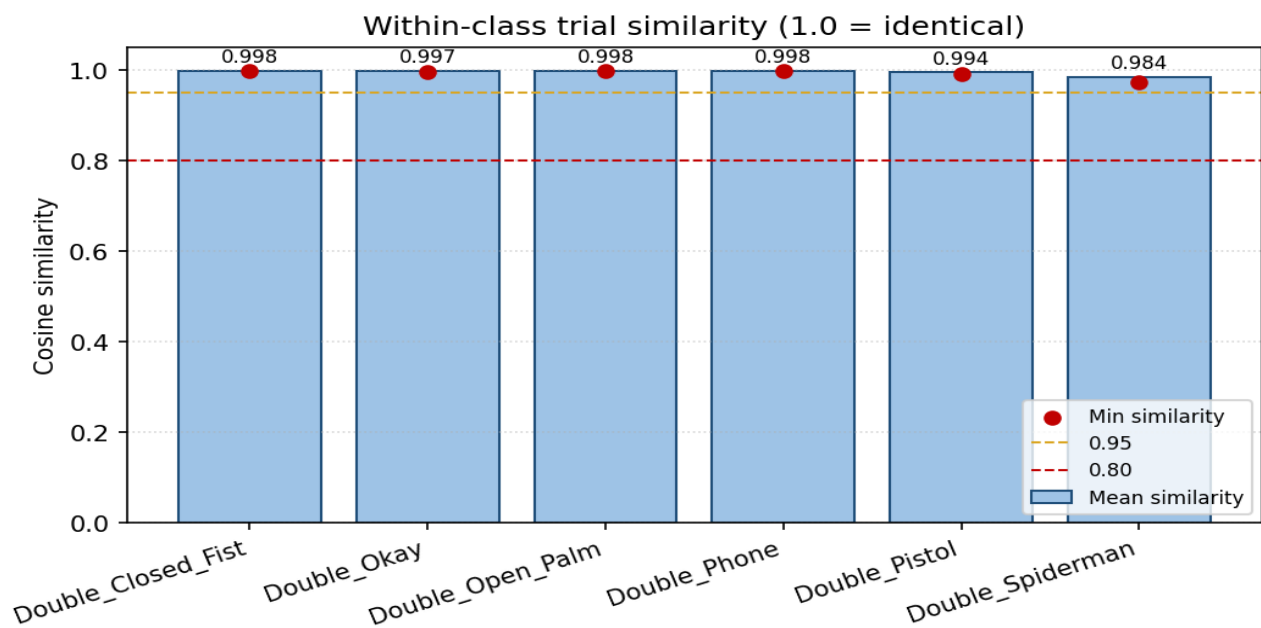


t-SNE feature space projection

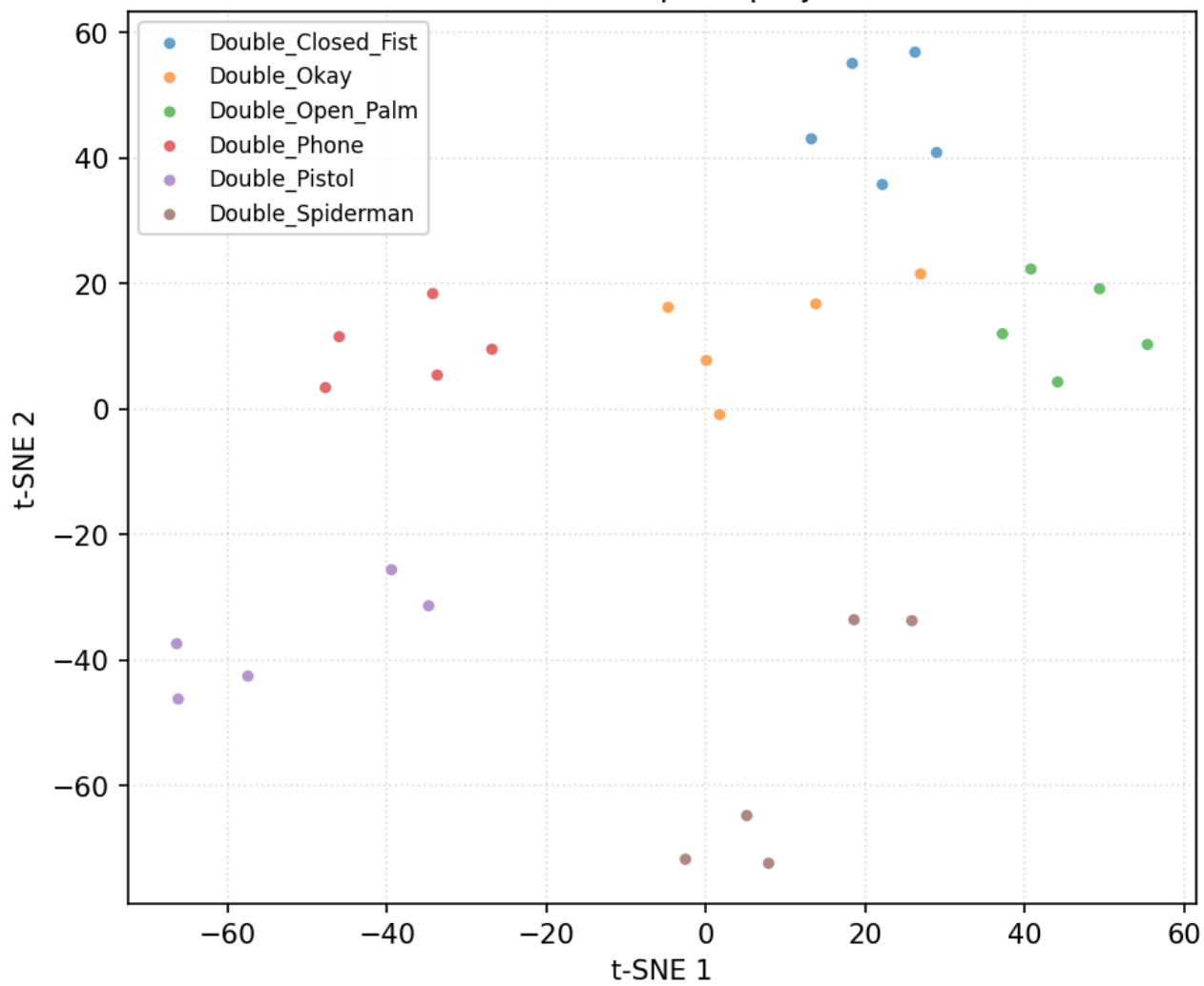


Data diagnostics — user 5_Georgia

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

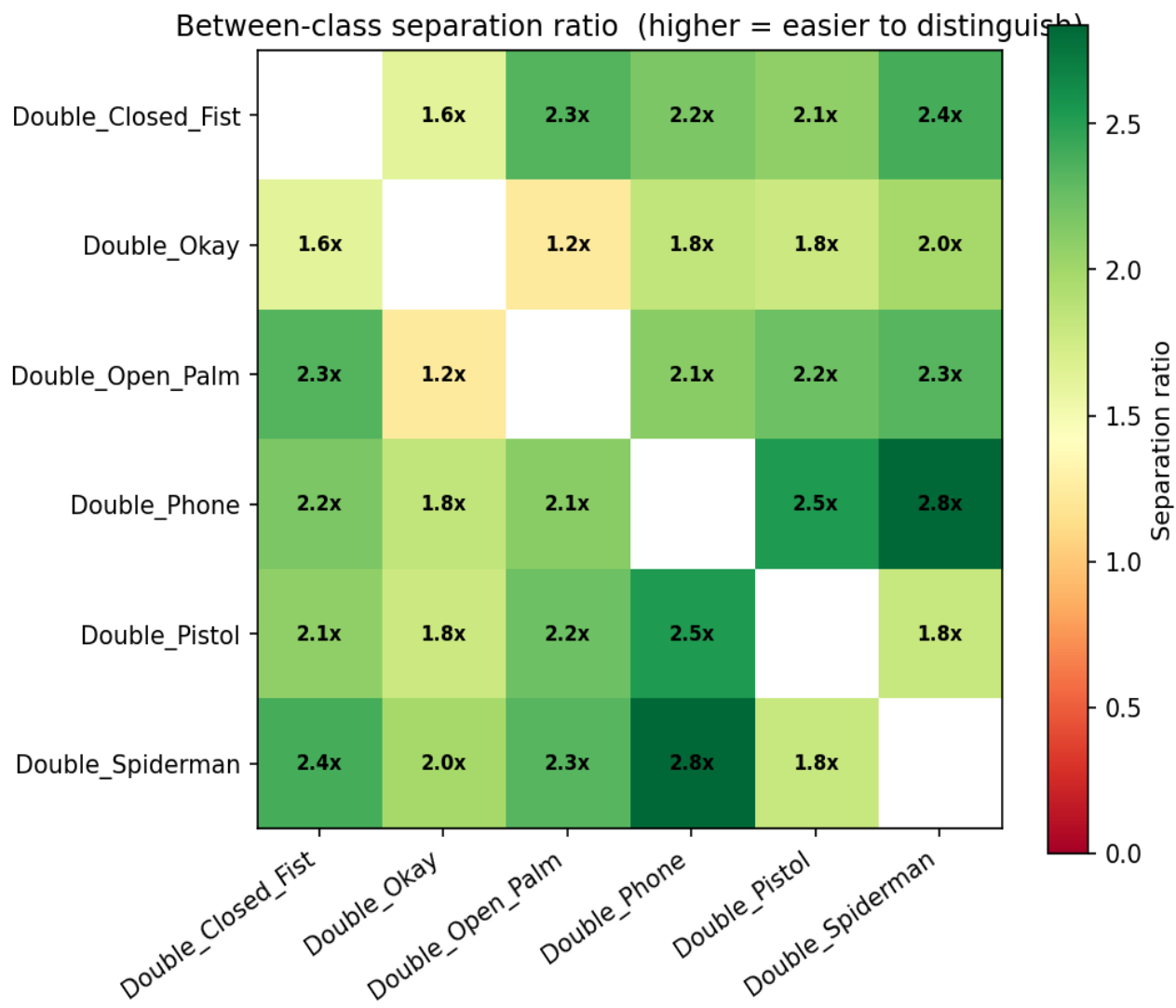
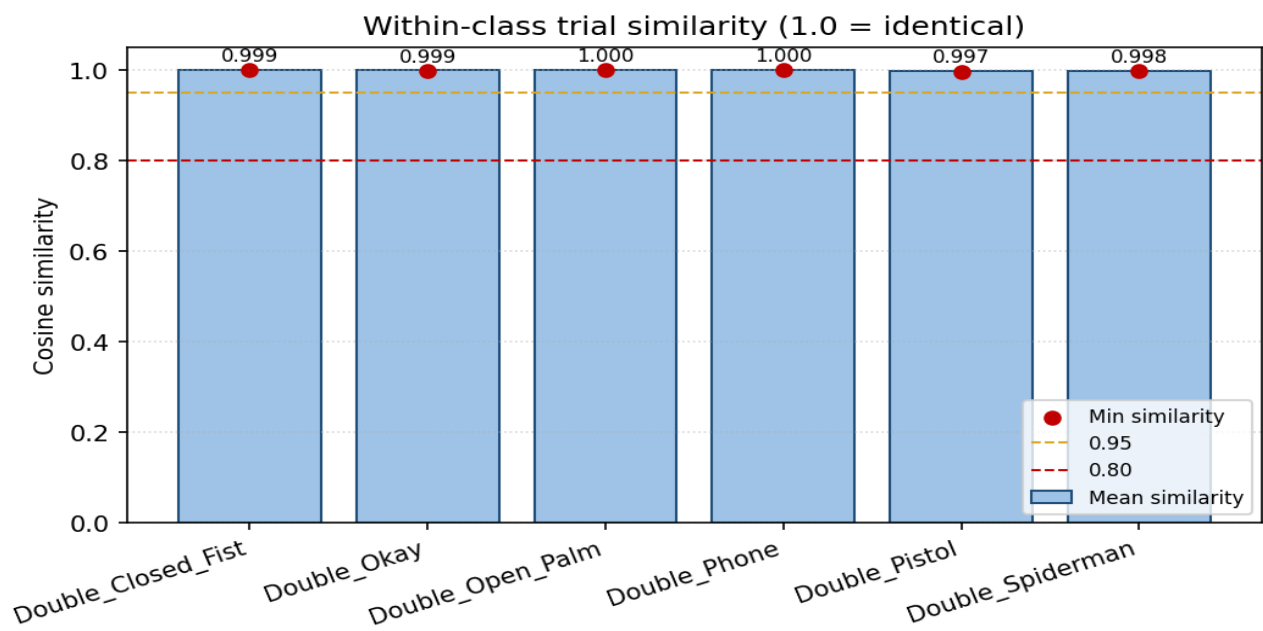


t-SNE feature space projection

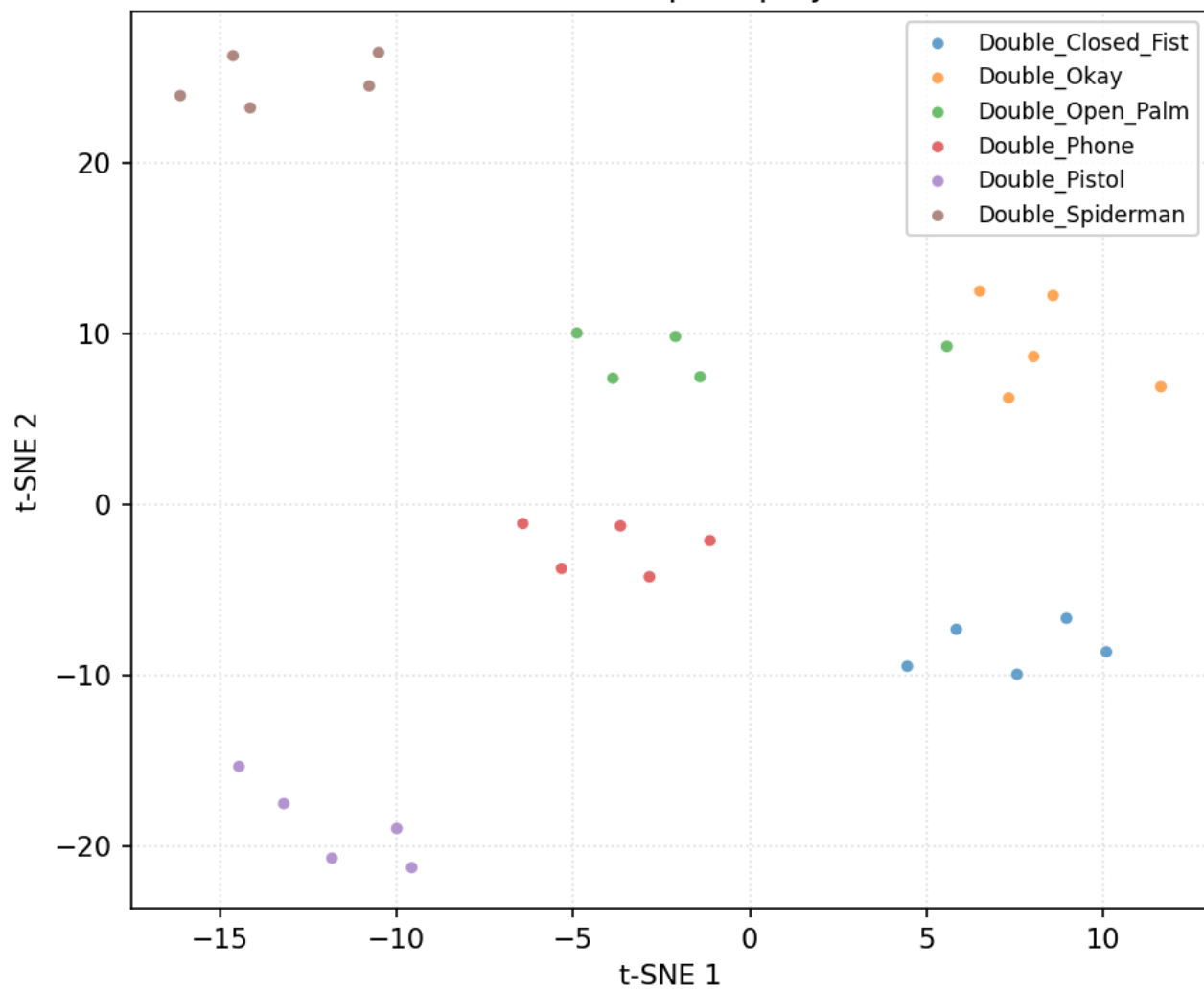


Data diagnostics — user 6_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

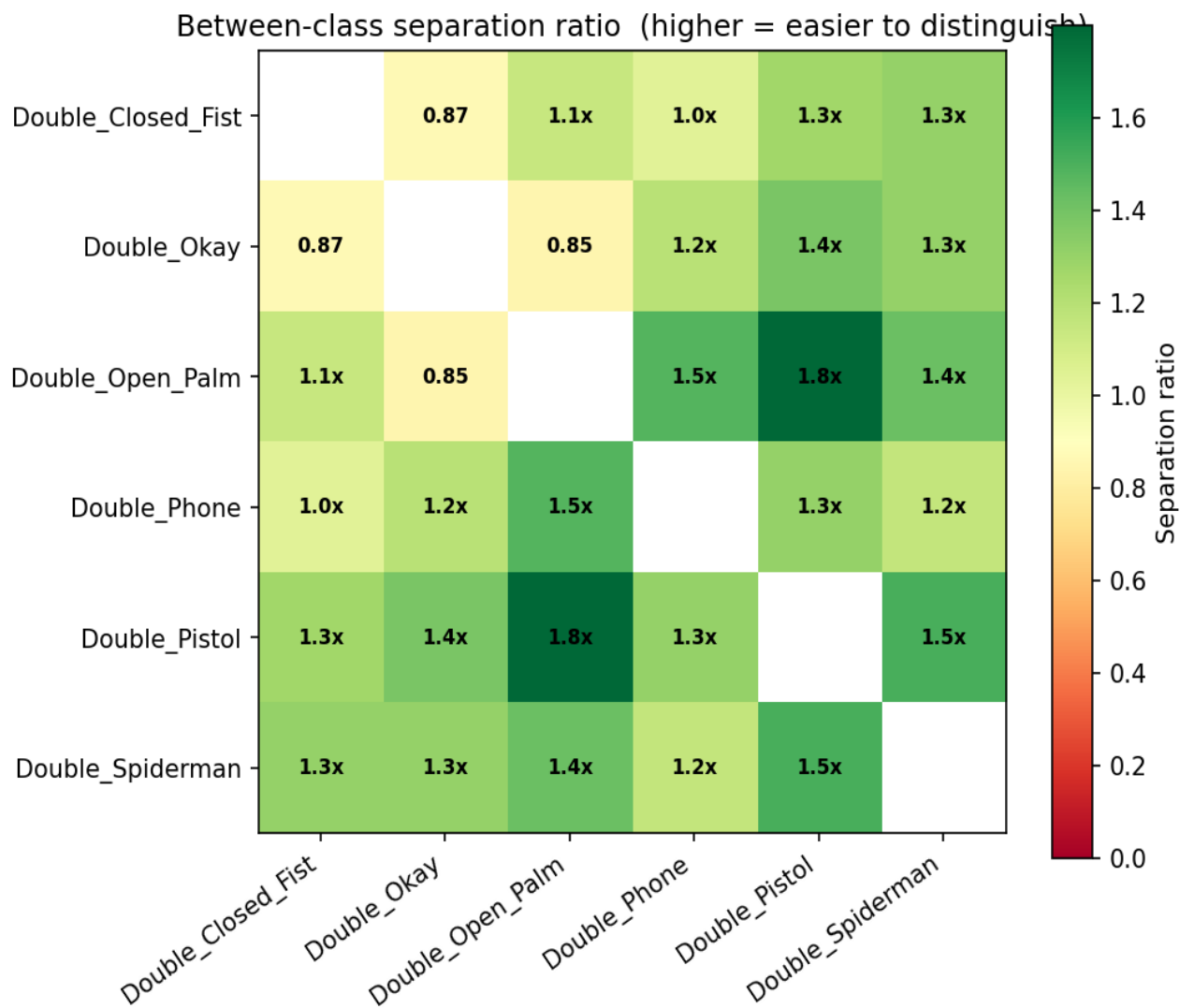
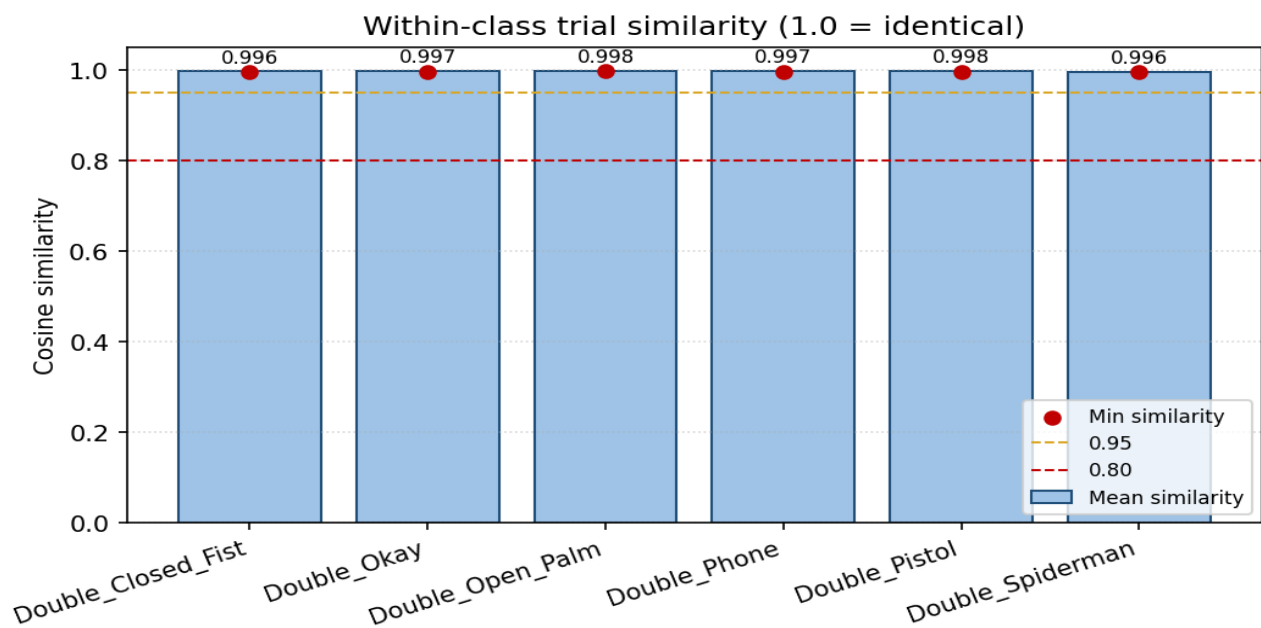


t-SNE feature space projection

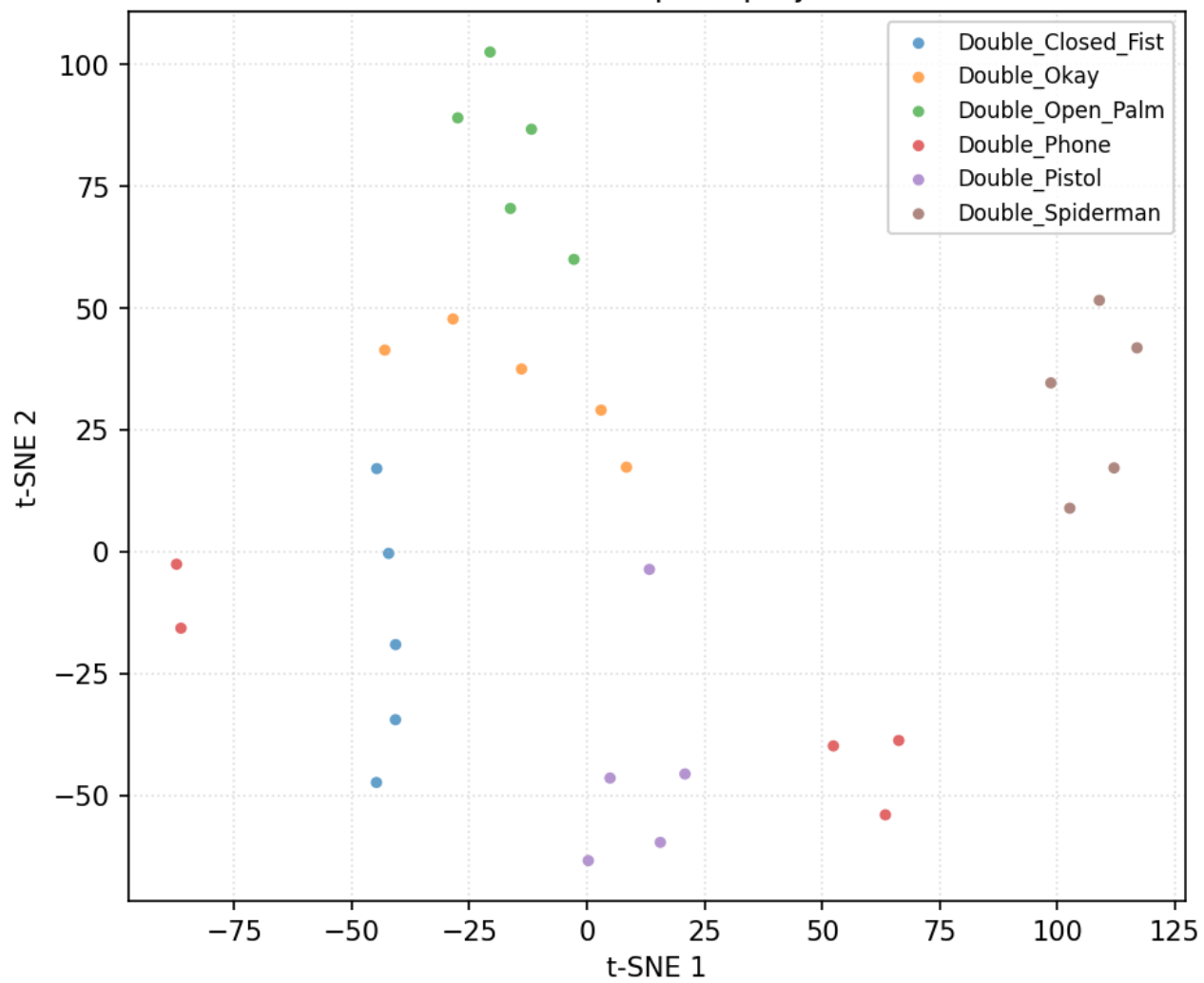


Data diagnostics — user 7_Henry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

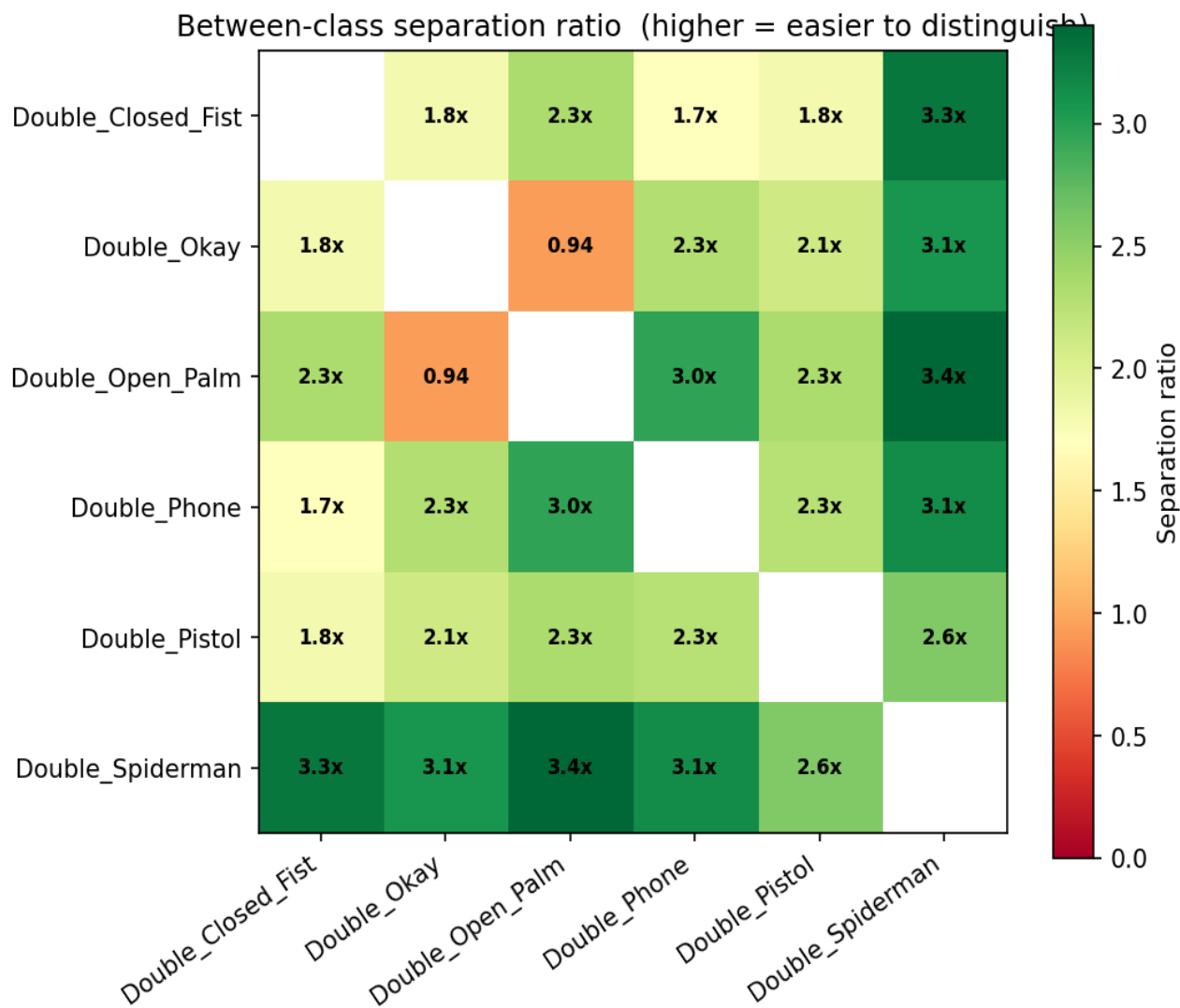
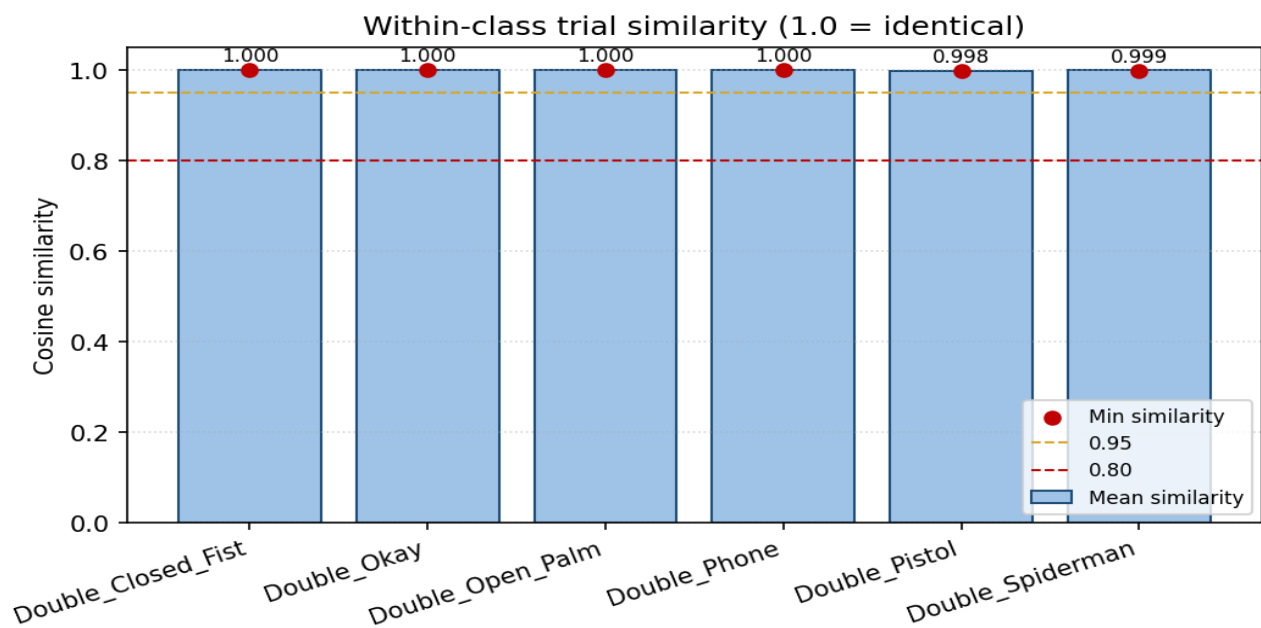


t-SNE feature space projection

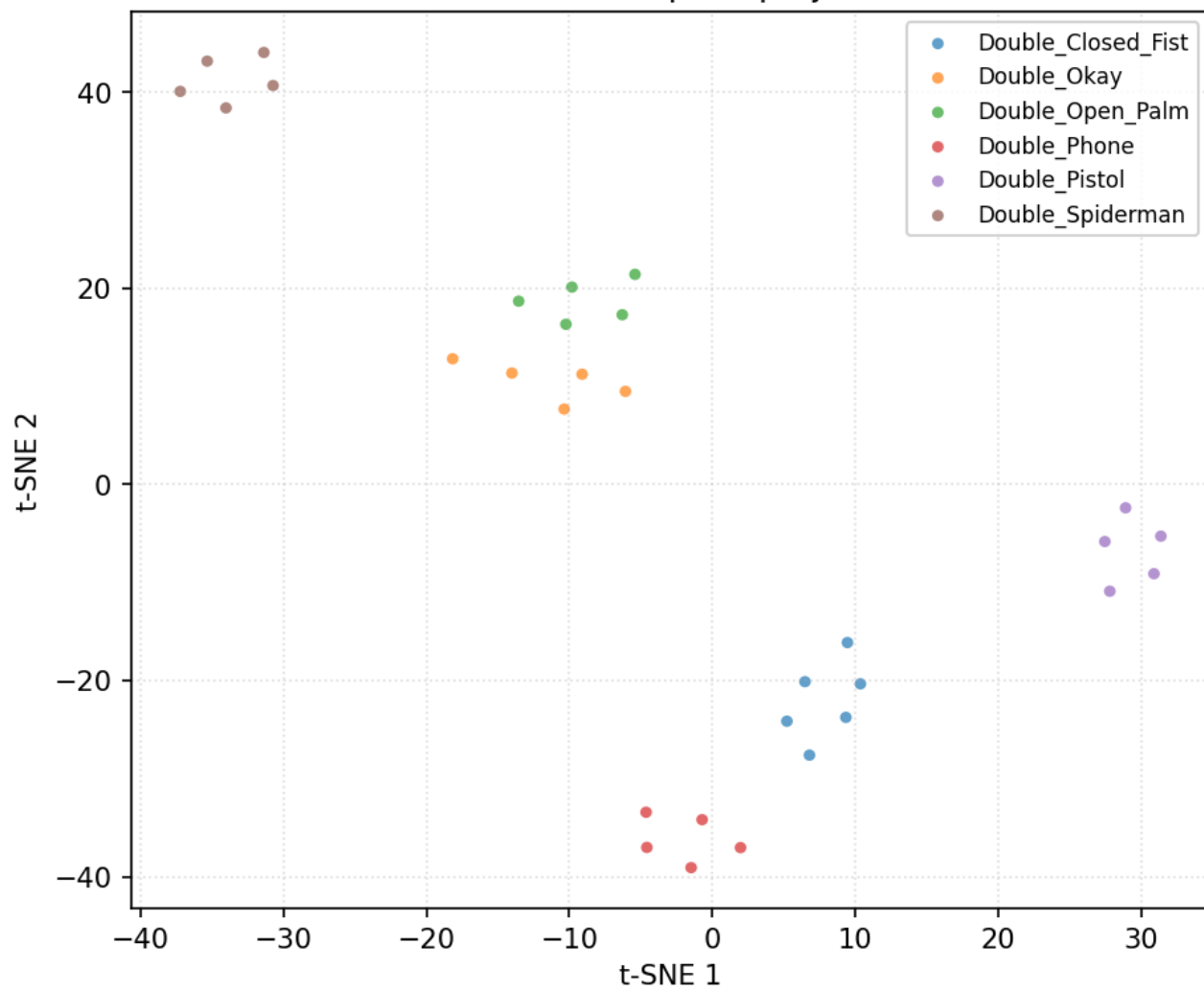


Data diagnostics — user 8_Jestin

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

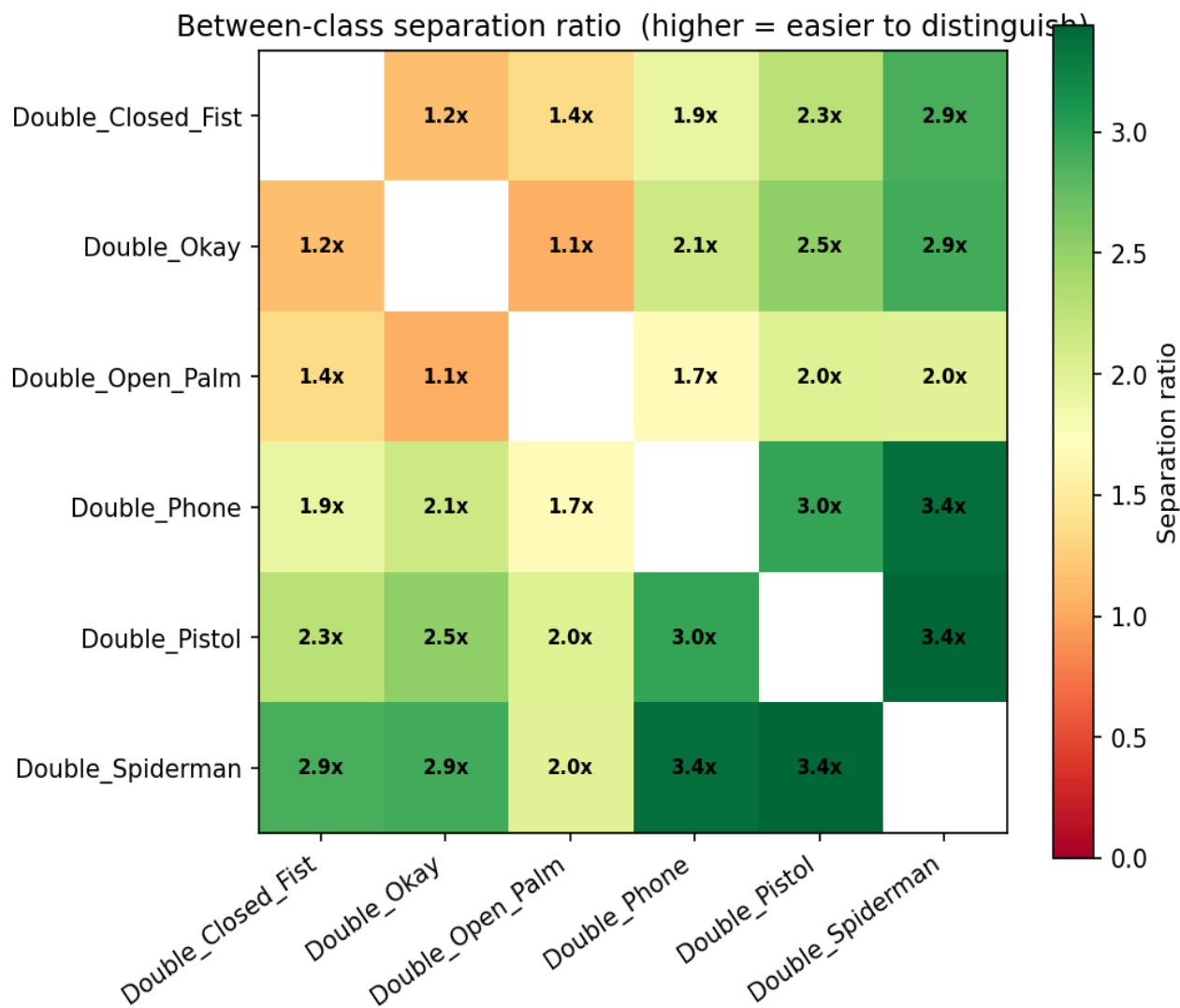
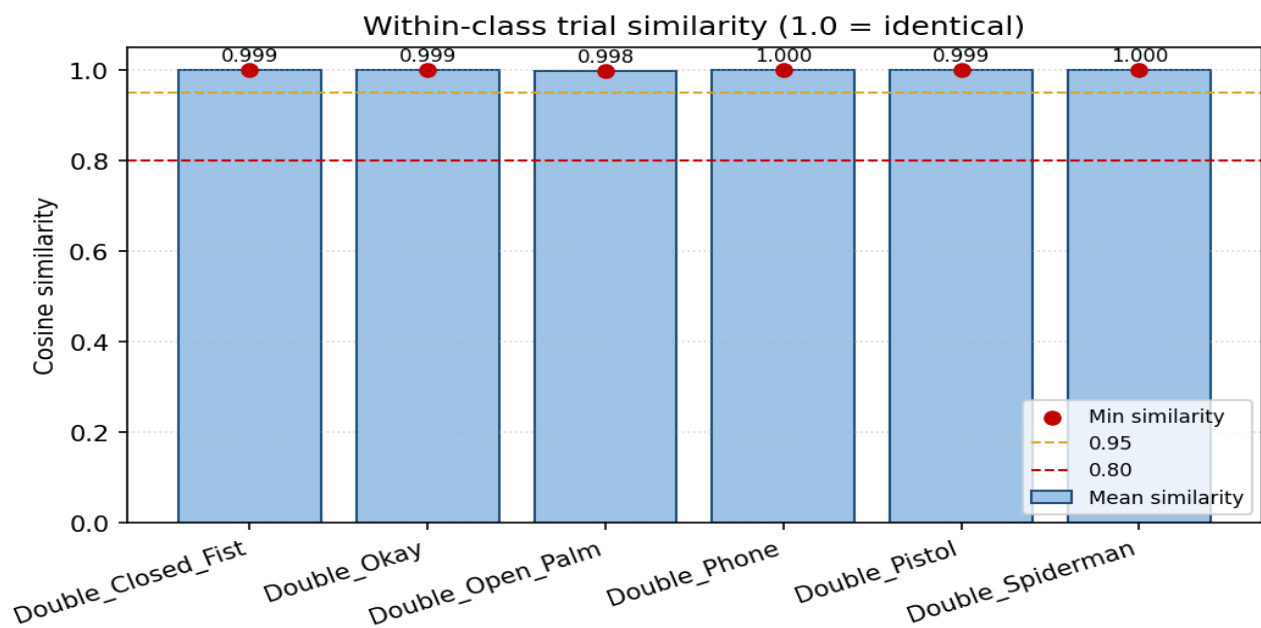


t-SNE feature space projection

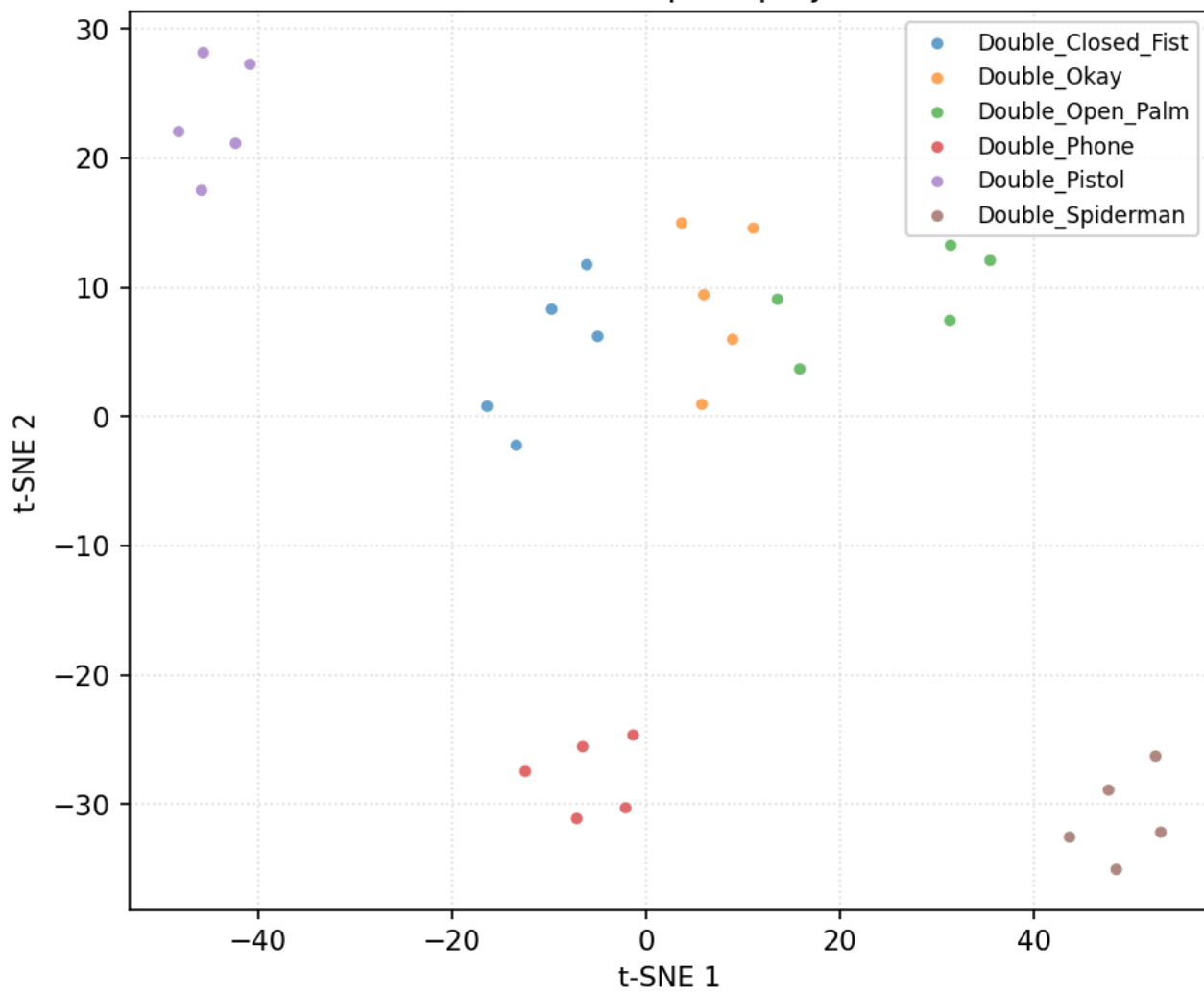


Data diagnostics — user 9_Joselyn

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: BN_GAP_Wide

Conv1D(32,k=8)+BN→Conv1D(64,k=8)+BN→Conv1D(128,k=5)+BN→GAP→Dropout(0.3)→Softmax [LOUO leader]

Trainable params	104,294
LOUO mean ± std	0.8493 ± 0.0834
Folds run	16
Inner-val mode	user
Per-trial norm	zscore

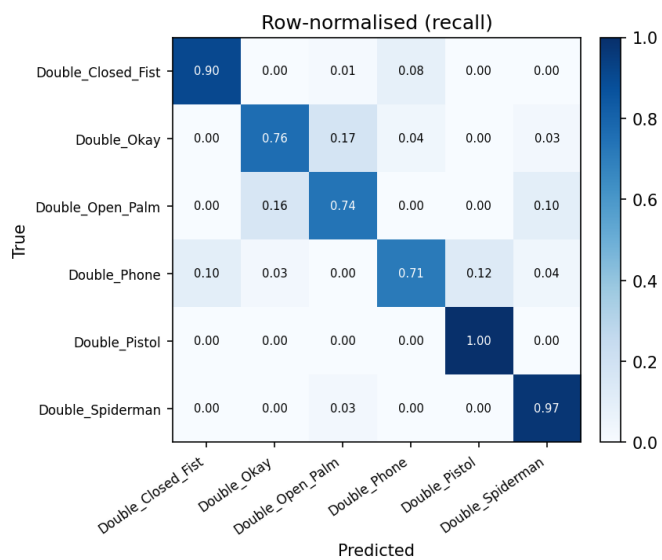
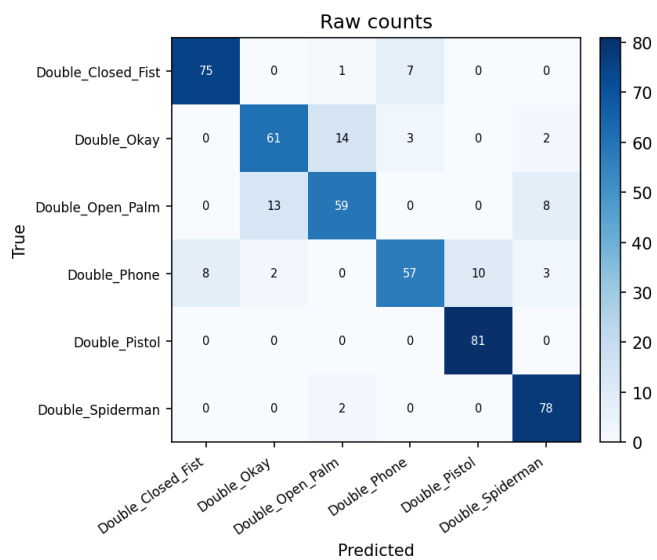
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	10_Josh	11_Mansh	423	30	0.9000
2	11_Mansh	12_Marcus	423	31	0.9355
3	12_Marcus	13_Nat	422	30	0.7667
4	13_Nat	14_StephenV2	422	32	0.7812
5	14_StephenV2	16_Bella	424	30	0.9333
6	16_Bella	17_Raquel	424	30	0.9333
7	17_Raquel	1_Alan	424	30	0.8000
8	1_Alan	2_Alex	424	30	0.8667
9	2_Alex	3_Anghad	424	30	0.6667
10	3_Anghad	4_Daniel	424	30	0.8333
11	4_Daniel	5_Georgia	424	30	1.0000
12	5_Georgia	6_Harry	424	30	0.8667
13	6_Harry	7_Henry	424	30	0.9000
14	7_Henry	8_Jestin	423	30	0.7333
15	8_Jestin	9_Joselyn	423	31	0.8387
16	9_Joselyn	10_Josh	424	30	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Closed_Fist	0.904	0.904	0.904	83
Double_Okay	0.803	0.762	0.782	80
Double_Open_Palm	0.776	0.738	0.756	80
Double_Phone	0.851	0.713	0.776	80
Double_Pistol	0.890	1.000	0.942	81
Double_Spiderman	0.857	0.975	0.912	80
macro avg	0.847	0.849	0.845	484
weighted avg	0.847	0.849	0.846	484

Confusion matrix (aggregated across folds)



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
LOUO mean ± std	0.8465 ± 0.0778
Folds run	16
Inner-val mode	user
Per-trial norm	zscore

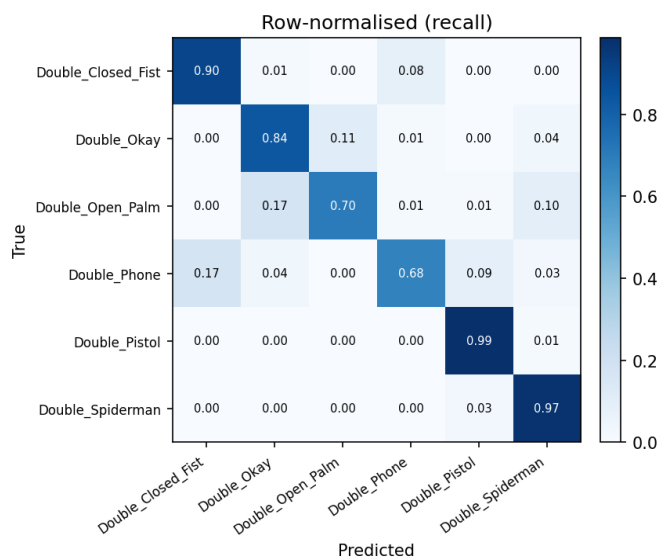
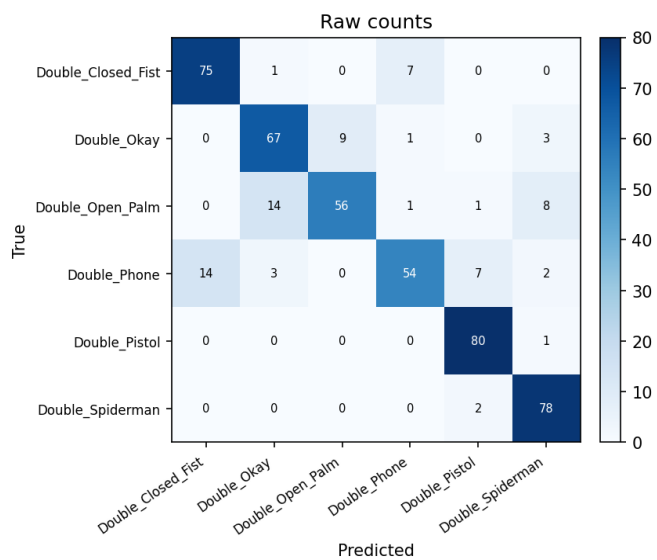
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	10_Josh	11_Mansh	423	30	0.8333
2	11_Mansh	12_Marcus	423	31	0.9355
3	12_Marcus	13_Nat	422	30	0.9667
4	13_Nat	14_StephenV2	422	32	0.9062
5	14_StephenV2	16_Bella	424	30	0.8000
6	16_Bella	17_Raquel	424	30	0.9000
7	17_Raquel	1_Alan	424	30	0.8333
8	1_Alan	2_Alex	424	30	0.8000
9	2_Alex	3_Anghad	424	30	0.7000
10	3_Anghad	4_Daniel	424	30	0.7333
11	4_Daniel	5_Georgia	424	30	0.9000
12	5_Georgia	6_Harry	424	30	0.8000
13	6_Harry	7_Henry	424	30	0.8667
14	7_Henry	8_Jestin	423	30	0.7333
15	8_Jestin	9_Joselyn	423	31	0.9355
16	9_Joselyn	10_Josh	424	30	0.9000

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Closed_Fist	0.843	0.904	0.872	83
Double_Okay	0.788	0.838	0.812	80
Double_Open_Palm	0.862	0.700	0.772	80
Double_Phone	0.857	0.675	0.755	80
Double_Pistol	0.889	0.988	0.936	81
Double_Spiderman	0.848	0.975	0.907	80
macro avg	0.848	0.846	0.842	484
weighted avg	0.848	0.847	0.843	484

Confusion matrix (aggregated across folds)



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	203,878
LOUO mean $\pm$ std	0.8255 $\pm$ 0.1215
Folds run	16
Inner-val mode	user
Per-trial norm	zscore

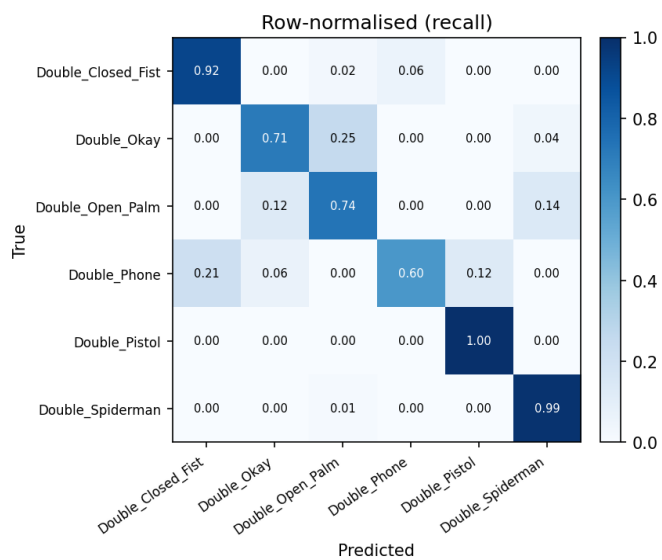
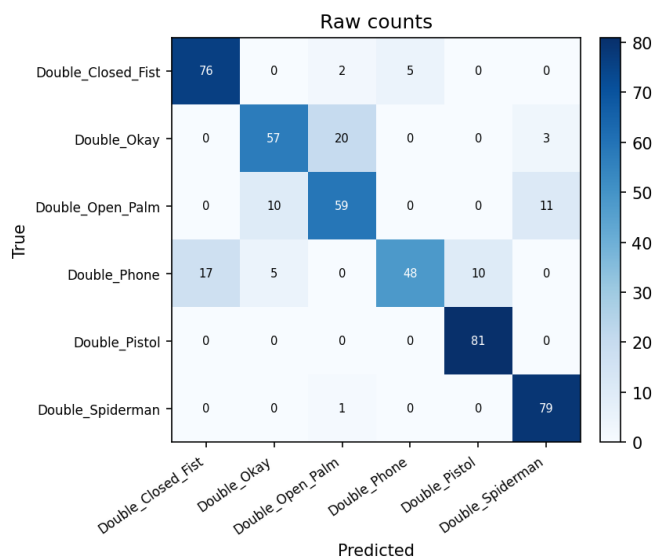
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	10_Josh	11_Mansh	423	30	0.8667
2	11_Mansh	12_Marcus	423	31	0.9677
3	12_Marcus	13_Nat	422	30	0.8333
4	13_Nat	14_StephenV2	422	32	0.9375
5	14_StephenV2	16_Bella	424	30	0.9000
6	16_Bella	17_Raquel	424	30	1.0000
7	17_Raquel	1_Alan	424	30	0.8000
8	1_Alan	2_Alex	424	30	0.8333
9	2_Alex	3_Anghad	424	30	0.5000
10	3_Anghad	4_Daniel	424	30	0.7333
11	4_Daniel	5_Georgia	424	30	0.7333
12	5_Georgia	6_Harry	424	30	0.7333
13	6_Harry	7_Henry	424	30	0.9333
14	7_Henry	8_Jestin	423	30	0.7000
15	8_Jestin	9_Joselyn	423	31	0.9032
16	9_Joselyn	10_Josh	424	30	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Closed_Fist	0.817	0.916	0.864	83
Double_Okay	0.792	0.713	0.750	80
Double_Open_Palm	0.720	0.738	0.728	80
Double_Phone	0.906	0.600	0.722	80
Double_Pistol	0.890	1.000	0.942	81
Double_Spiderman	0.849	0.988	0.913	80
macro avg	0.829	0.826	0.820	484
weighted avg	0.829	0.826	0.820	484

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,190
LOUO mean $\pm$ std	0.7994 $\pm$ 0.1147
Folds run	16
Inner-val mode	user
Per-trial norm	zscore

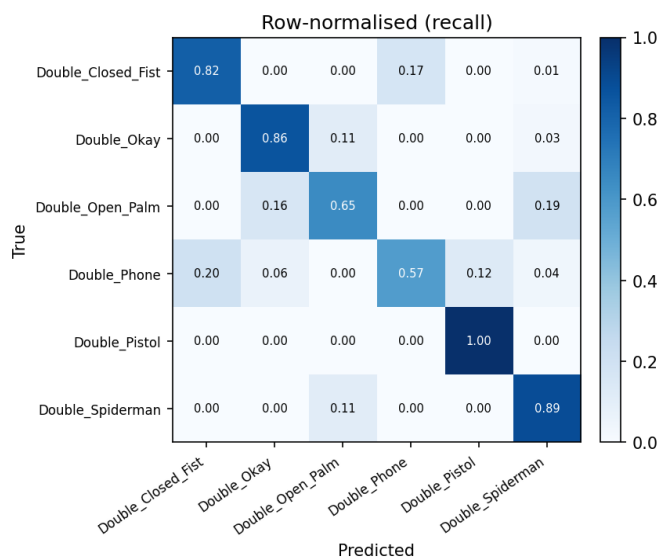
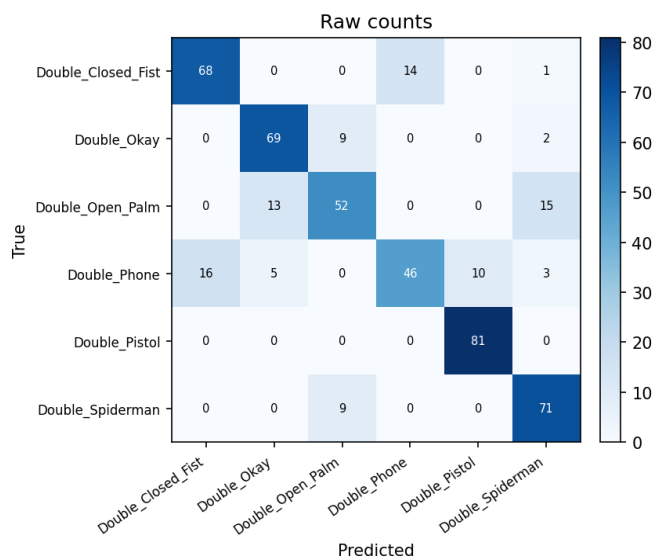
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	10_Josh	11_Mansh	423	30	0.9333
2	11_Mansh	12_Marcus	423	31	0.8065
3	12_Marcus	13_Nat	422	30	0.9000
4	13_Nat	14_StephenV2	422	32	0.8125
5	14_StephenV2	16_Bella	424	30	0.7667
6	16_Bella	17_Raquel	424	30	0.8000
7	17_Raquel	1_Alan	424	30	0.8333
8	1_Alan	2_Alex	424	30	0.7667
9	2_Alex	3_Anghad	424	30	0.4333
10	3_Anghad	4_Daniel	424	30	0.8333
11	4_Daniel	5_Georgia	424	30	0.8667
12	5_Georgia	6_Harry	424	30	0.8667
13	6_Harry	7_Henry	424	30	0.8333
14	7_Henry	8_Jestin	423	30	0.6333
15	8_Jestin	9_Joselyn	423	31	0.8387
16	9_Joselyn	10_Josh	424	30	0.8667

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Closed_Fist	0.810	0.819	0.814	83
Double_Okay	0.793	0.863	0.826	80
Double_Open_Palm	0.743	0.650	0.693	80
Double_Phone	0.767	0.575	0.657	80
Double_Pistol	0.890	1.000	0.942	81
Double_Spiderman	0.772	0.887	0.826	80
macro avg	0.796	0.799	0.793	484
weighted avg	0.796	0.800	0.794	484

Confusion matrix (aggregated across folds)



Final retrain (deployment model)

Variant	BN_GAP_Wide
Held-out user	— (trained on all users)
Held-out accuracy	—
Training-set size	484 samples (484 before augmentation)
Trainable params	104,294
Final train loss	0.0207
Final train accuracy	1.0000
Model path	— (not saved)
Labels path	— (not saved)
Scaler path	— (per-trial norm in use)